Introduction to Orthographic Projection

Engineering Drawing

Projections Explained…

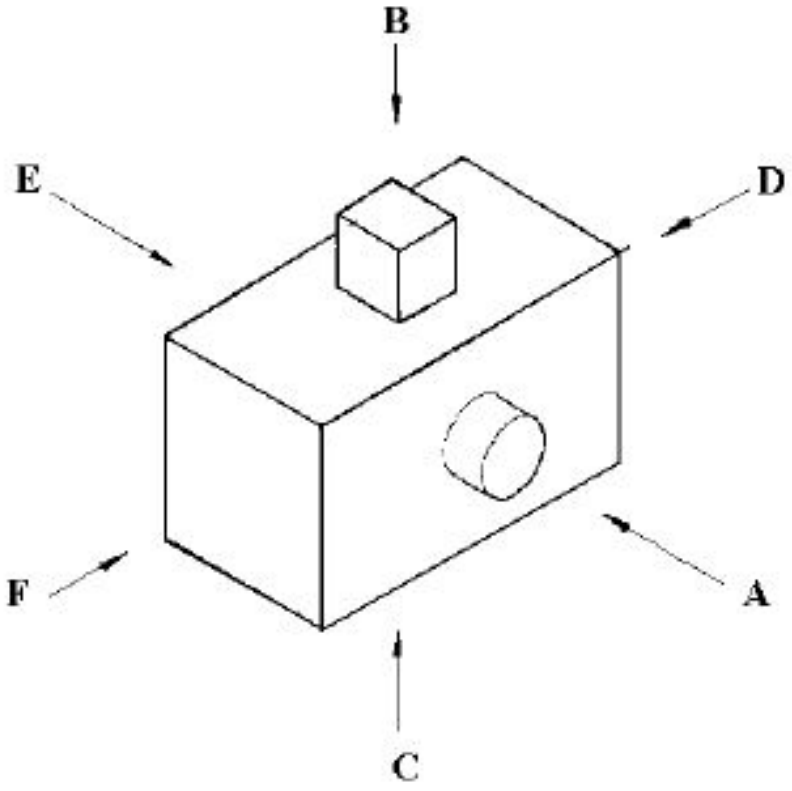
Topic	What students need to learn:
3 Processes, techniques and specialist tools continued	3.3 – Use of media to convey design decisions, to record to recognised standards, explain and communicate information and ideas using the following methods and techniques: a) Pictorial drawing methods for representing 3D forms – isometric, 2-point perspective b) Working drawings for communicating 2D technical information – 3rd angle orthographic projection, triangulation c) Nets (developments) for communicating information about 3D forms in a 2D format d) Translation between working drawings, pictorial drawings and nets (developments) e) Report writing.

Orthographic Projection

Starter *(5min – discussion)*

Take a look at the drawing on the RHS…

- *Is it constructed in 2D or 3D?*
- *What does the 'D' stand for when we are talking about 2D and 3D drawings?*
- *Why would it be foolish to ask someone to **make** this part using this drawing?*



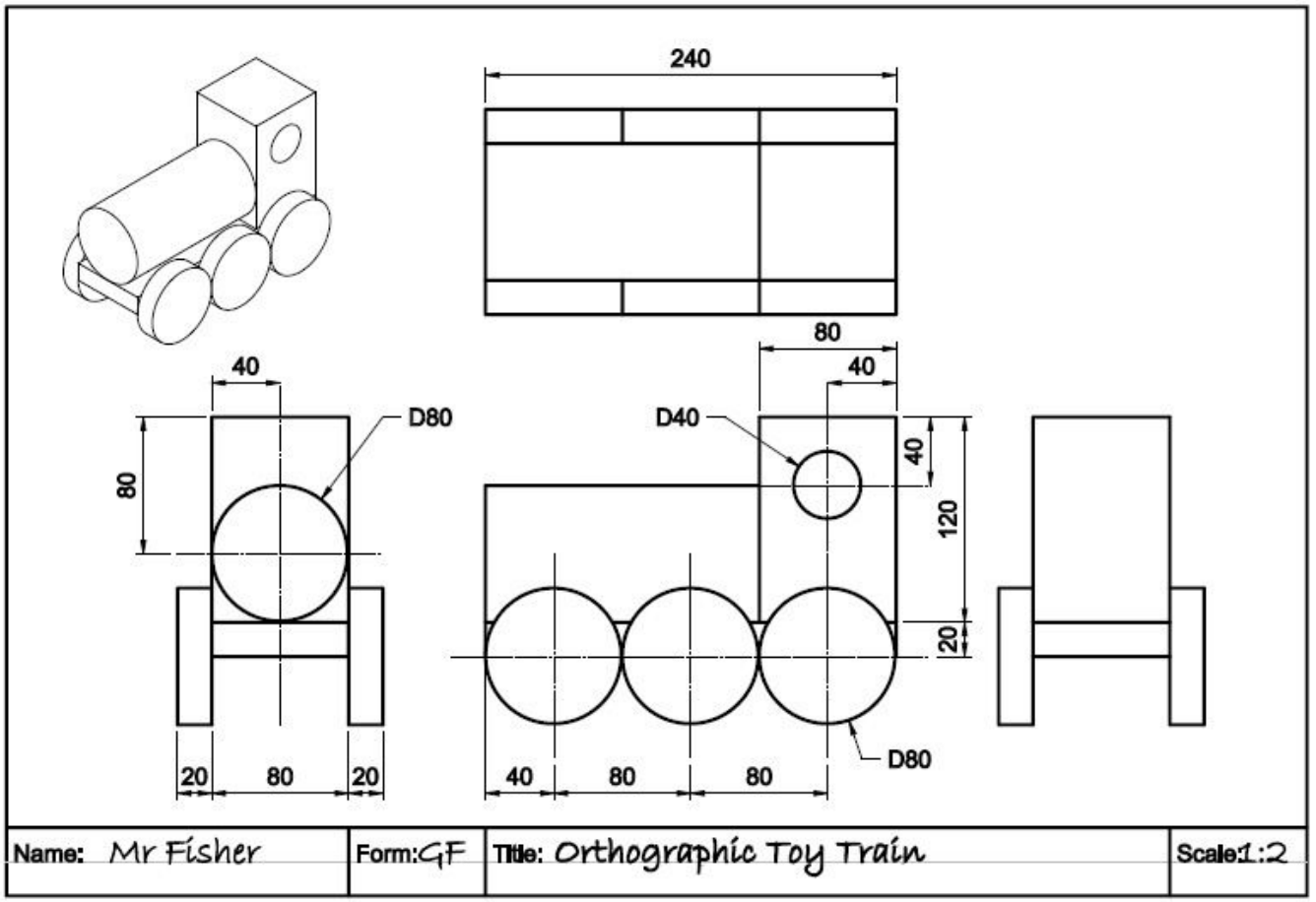
Learning Objectives

- To understand the need for *Orthographic Drawings*
- To be able to *read Orthographic Drawings*
- To be able to *construct Orthographic Drawings* – using basic drawing equipment

Learning Outcomes

- **ALL** students to develop a basic understanding of orthographic drawings
- **ALL** students to understand that an orthographic drawing consists of three views
- **MOST** students to understand that an orthographic drawing consists of a front view, plan view and side view
- **SOME** students to be able to construct three orthographic views from a dimensioned isometric drawing

An example of a ‘working drawing’…



A Quick History Lesson

Orthographic Projection

Orthographic Projection

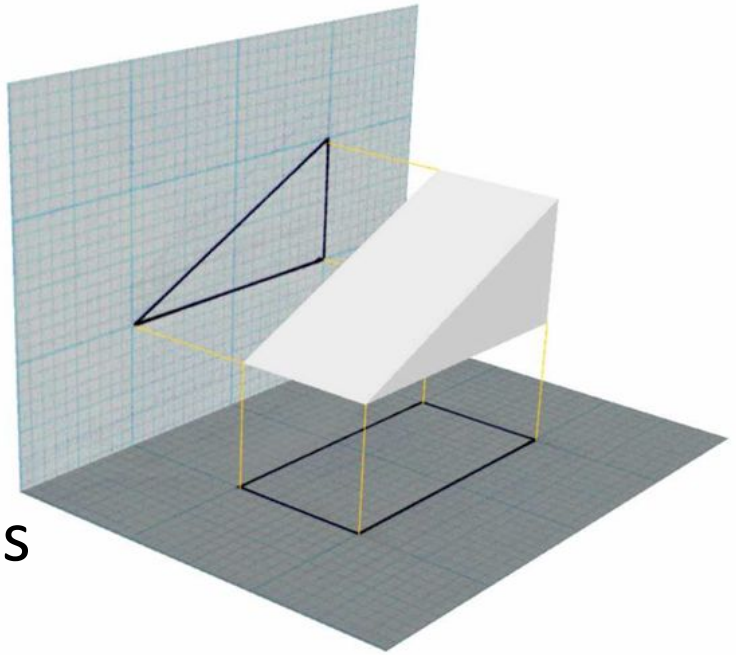
Back in the 18th century a French mathematician and engineer, Gaspard Monge (1746-1818), was involved with the design of military armoury. He developed a system, using two planes of projection at right angles to each other, for the graphical description of solid objects.



Descriptive Geometry

This system, which is called ***Descriptive Geometry***, provided a method of graphically describing objects accurately and unambiguously. It relied on ***the perpendicular projection of geometry from perpendicular planes***.

Monge's *Descriptive Geometry* forms the basis of what is now called ***Orthographic Projection***.



Two right angle planes of projection

Key Word:

The word ***orthographic*** means to *draw at right angles* and is derived from the Greek words:

ORTHOS - straight, rectangular, upright
GRAPHOS - written, drawn

Orthographic Projection

Orthographic projection is the graphical method used in modern engineering drawing. In order to interpret and communicate with engineering drawings a designer must have a sound understanding of it's use and a clear vision of how the various projections are created.

Orthographic Projection

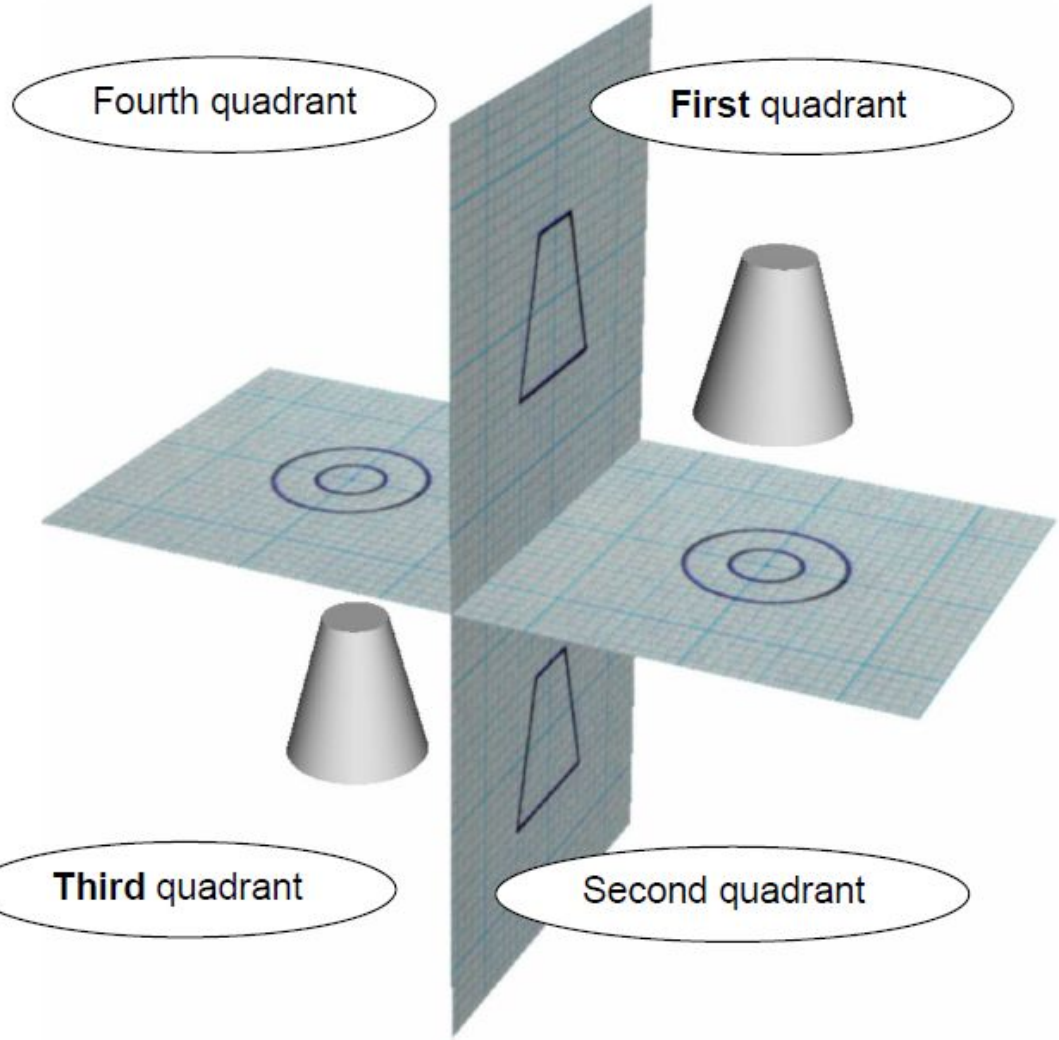
There are ***two*** predominant *orthographic projections* used today. They are based on Monge's original right angle planes and are shown fully in the next slide. They define *four separate spaces*, or *quadrants*. Each of these *quadrants* could contain the object to be represented. Traditionally however, only ***two*** are commonly used, ***the first*** and ***the third***.

Orthographic Drawing

Projections created with the ***object*** placed in the ***first quadrant***.

This means that the ***Vertical Plane*** *is* ***behind*** *the* ***object*** *and the* ***Horizontal Plane*** *is* ***underneath*** *the* ***object***.

This is known as ***First Angle projection***

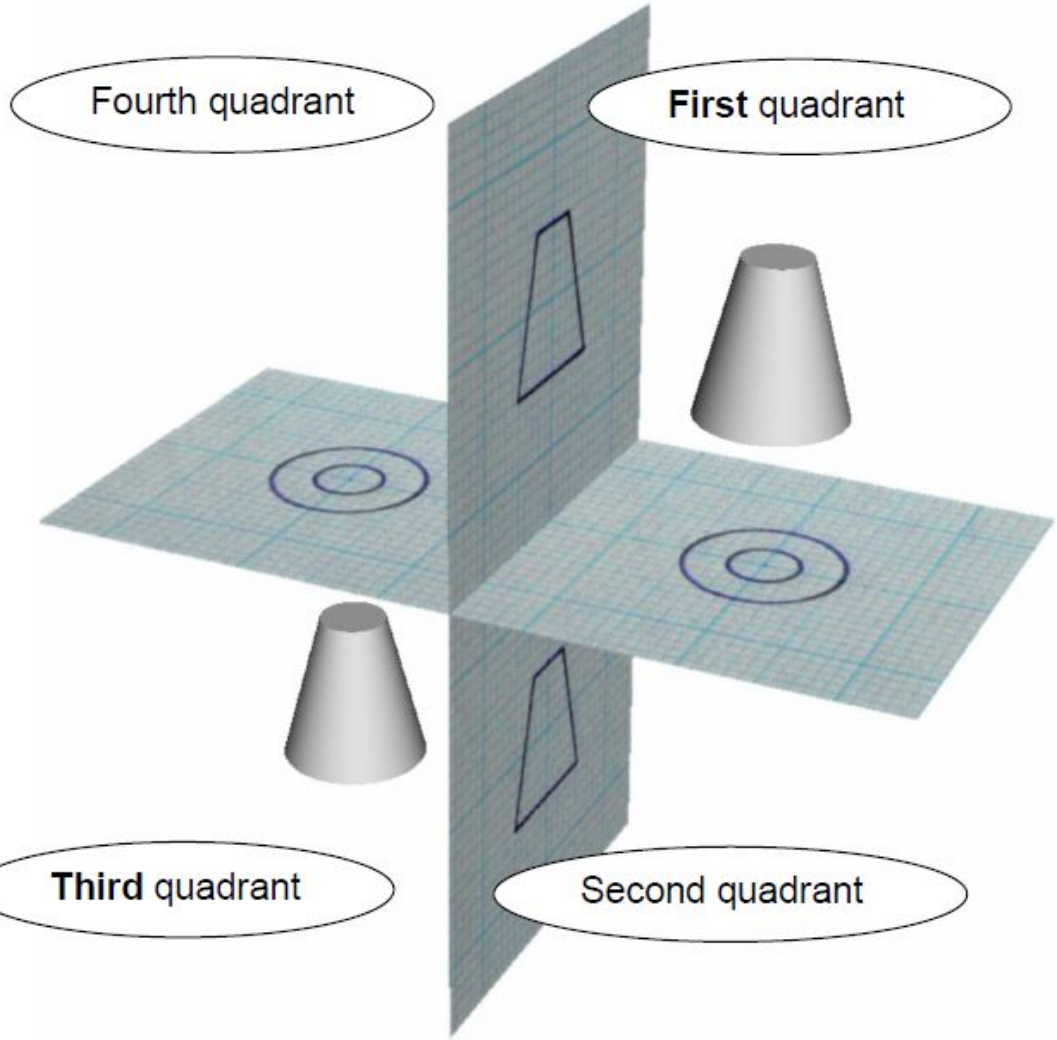


Orthographic Drawing

Projections created with the ***object*** placed in the ***third quadrant***.

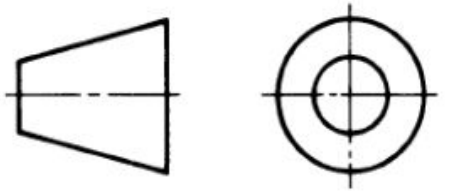
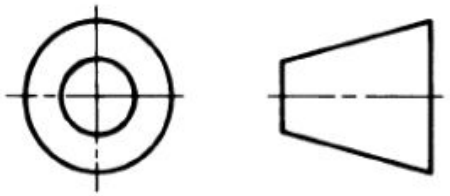
This means that the ***Vertical Plane*** *is in* ***front*** *of the* ***object*** *and the* ***Horizontal Plane*** *is* ***above*** *the* ***object*****.**

This is known as ***Third Angle projection***



Orthographic Projection Symbols

Both systems of projection, *First and Third angle*, are approved internationally and have equal status. The system used *must be clearly indicated on every drawing*, using the *appropriate symbol*

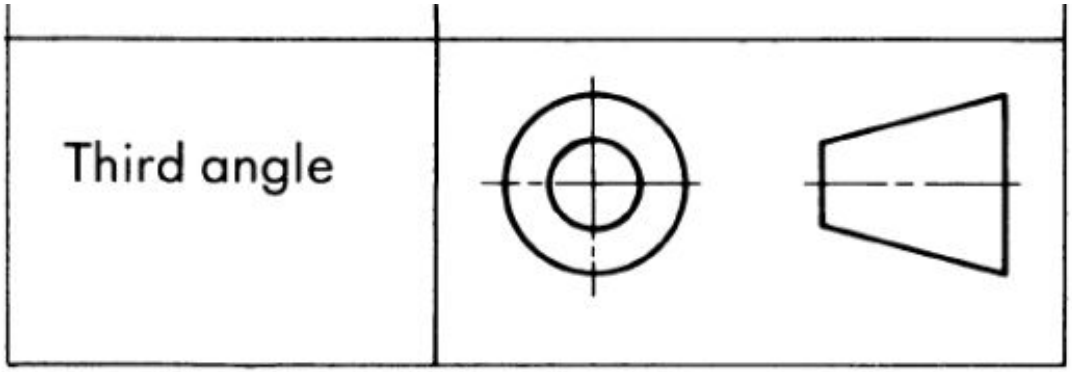
Projection	Symbol
First angle	
Third angle	

First Angle Projection – is more common in Europe

Third Angle Projection – is widely used both in the UK and America

Third Angle Projection

We are going to stick to third angle projection because it is easier to understand…



Orthographic Projection

Engineering Drawing

Working with Engineering Drawings…

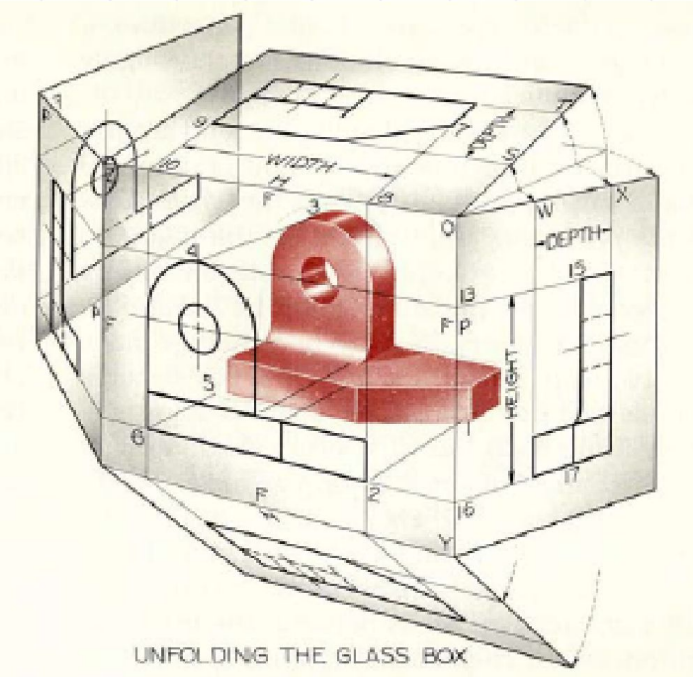
Orthographic Projection

Explained….

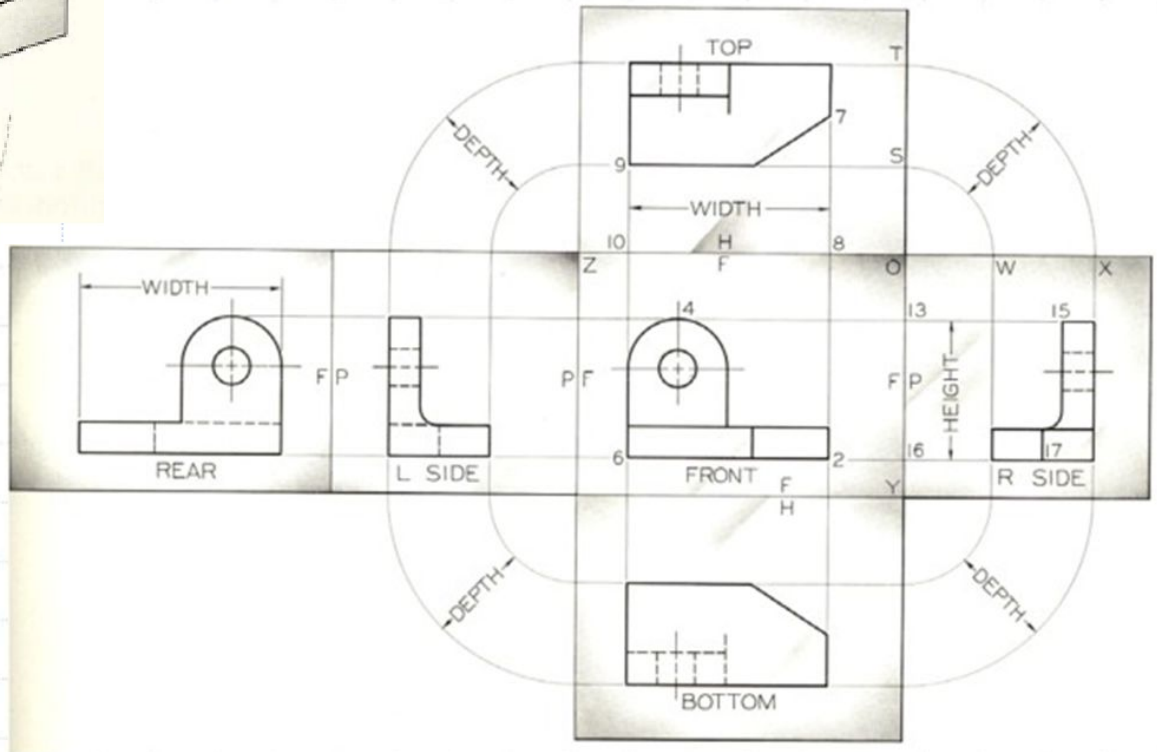
The Glass Box approach:

- Place a *3D object* inside a *glass box*
- Then *project parallel planes* from the ***object*** to the *principal planes forming the glass box* (always observed from outside the box)
- To show views of a *3D object* on a *2D piece of paper*, it is necessary to ***unfold*** the *planes* such that they lie in the *same plane*
- All *planes* except the *rear plane* are ***hinged*** to the ***frontal plane***, which is ***hinged*** to the ***left-side plane***

Glass Box Approach



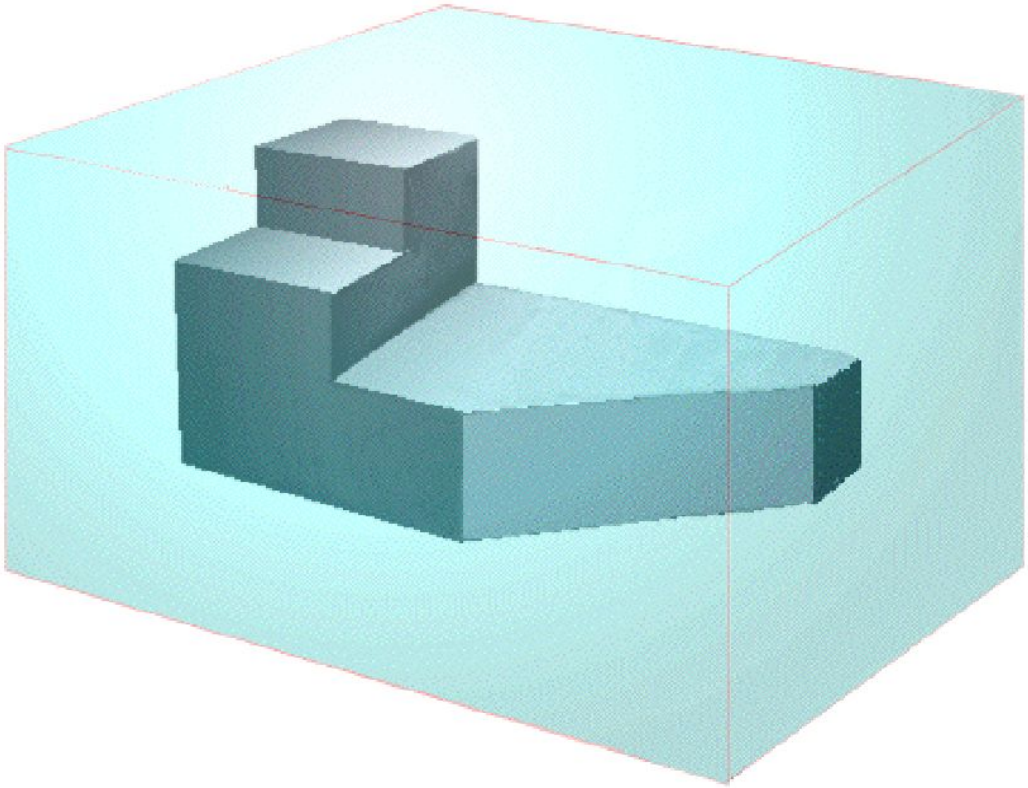
By *unfolding* the *glass box* the ***six views*** of the *object* can be seen



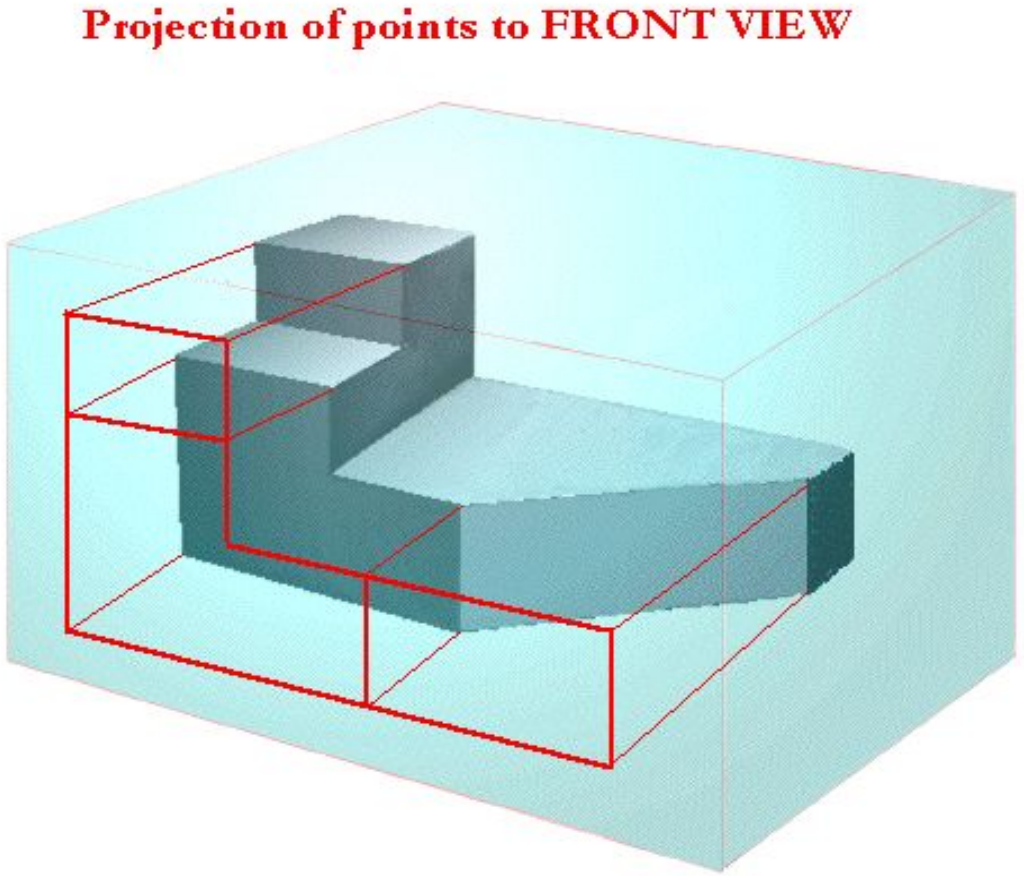
Glass Box Approach

Just to be clear…

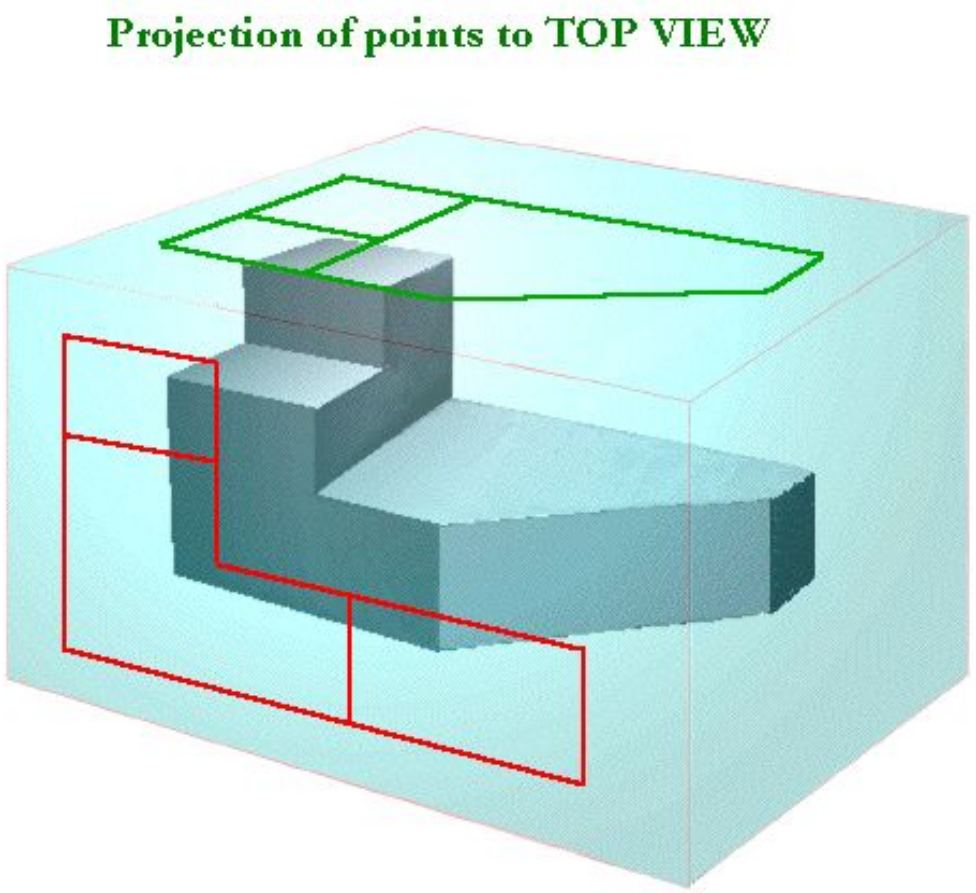
Projection of points to the three views



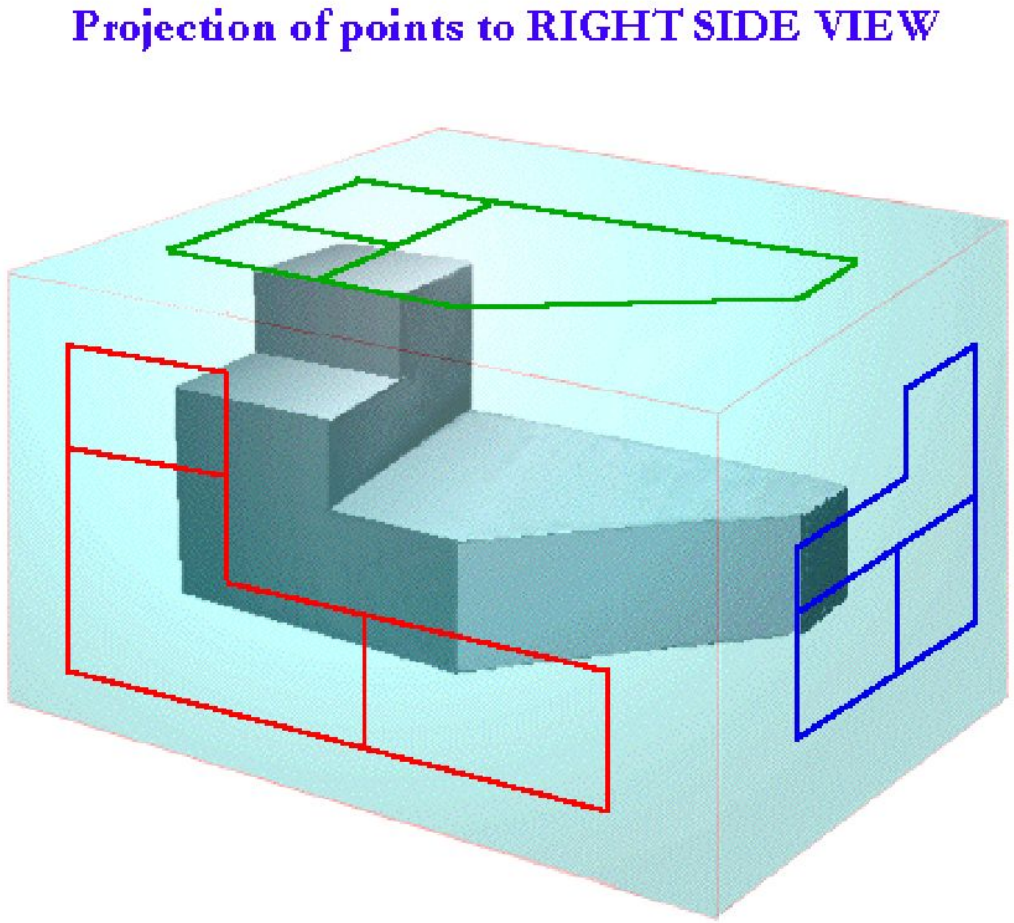
Glass Box Approach



Glass Box Approach

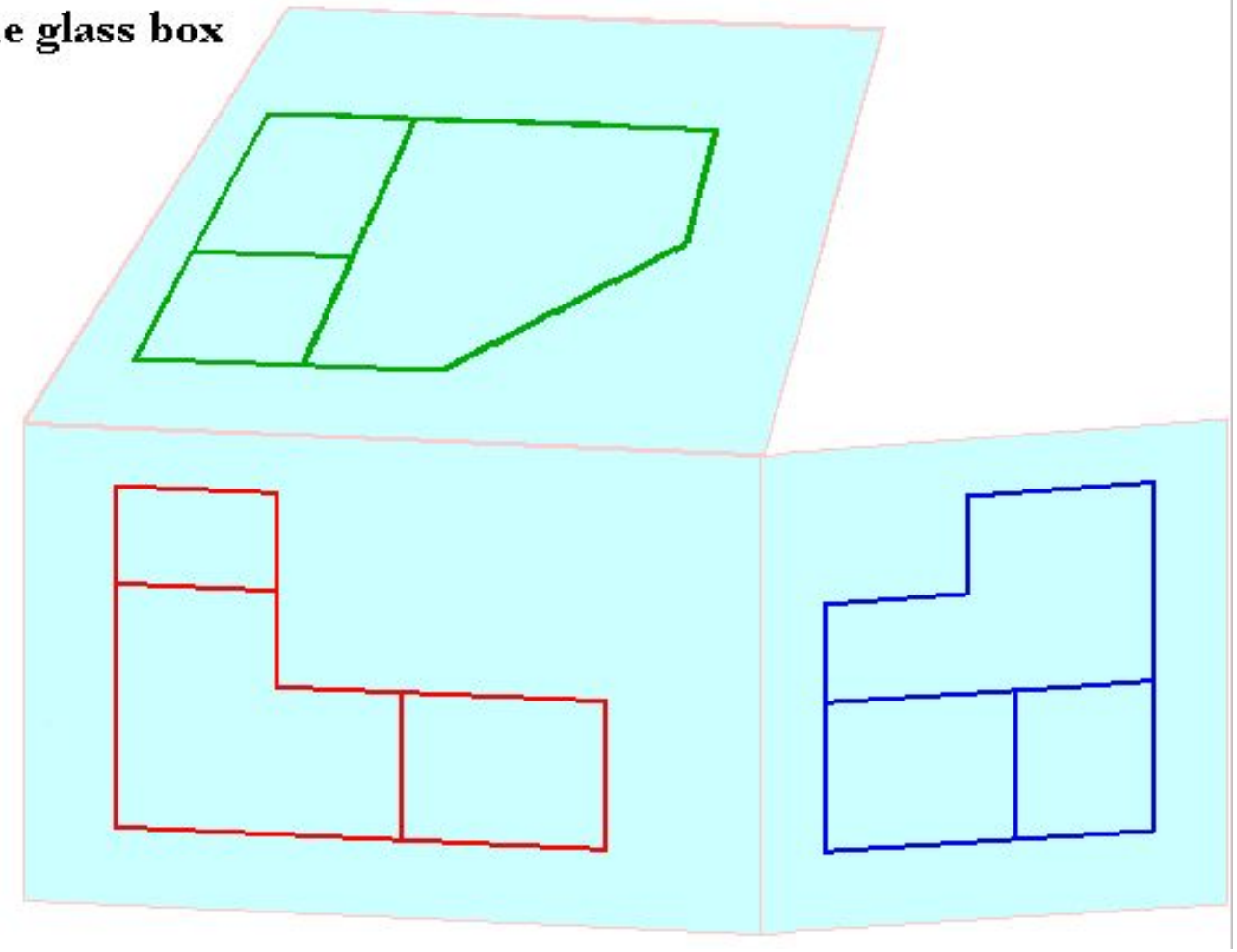


Glass Box Approach



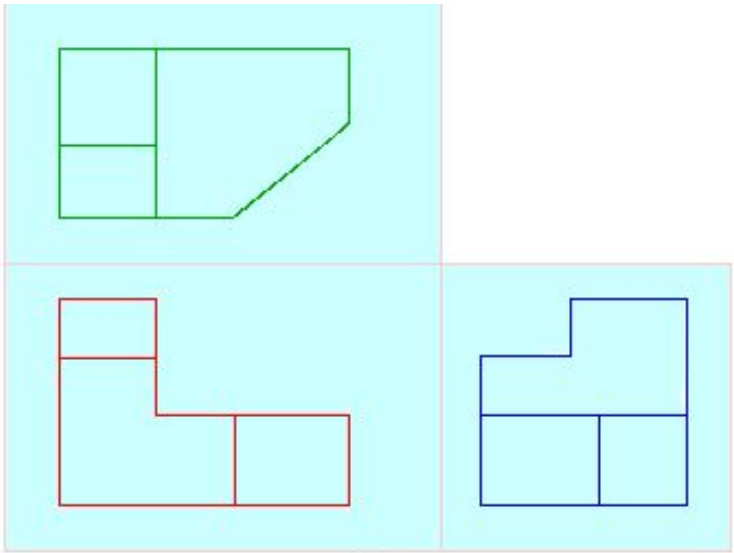
Glass Box Approach

Unfold the glass box

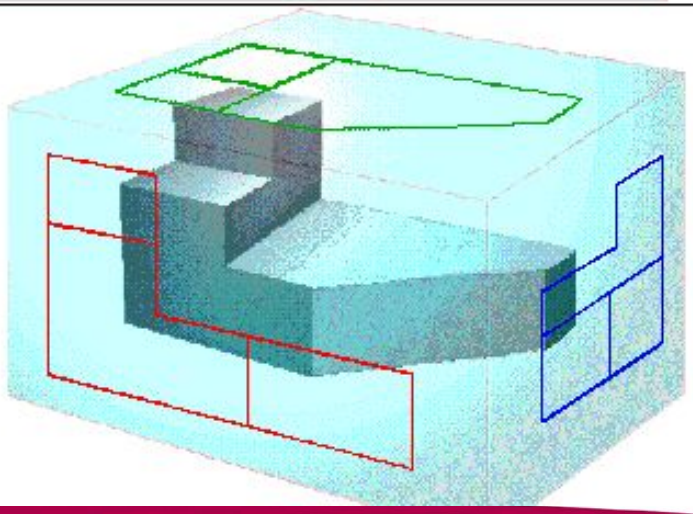


Glass Box Approach

Unfolded glass-box



Object in the glass-box



Orthographic Projection

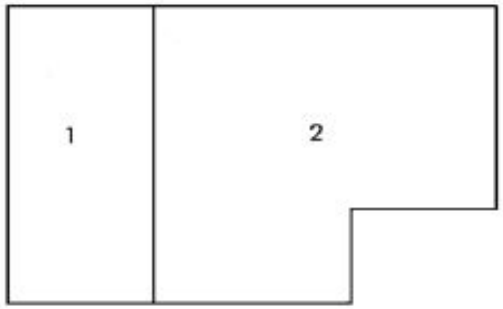
Activity: *(5 mins)*

Ok… have a look at these exercises…

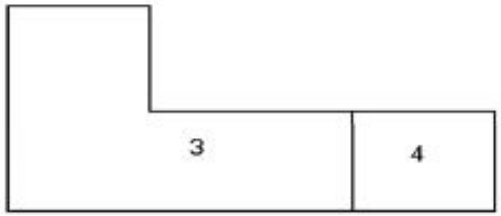
- ***Exercise 1-2-1***
- ***Exercise 1-2-2***

Exercise 1-2-1

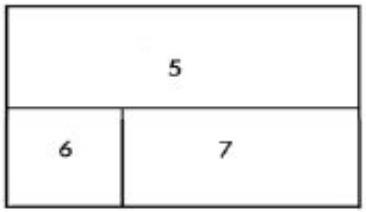
EXERCISES. Study the two drawings and complete the table by matching the numbered surfaces of the orthogonal drawing with the lettered surfaces of the isometric drawing.



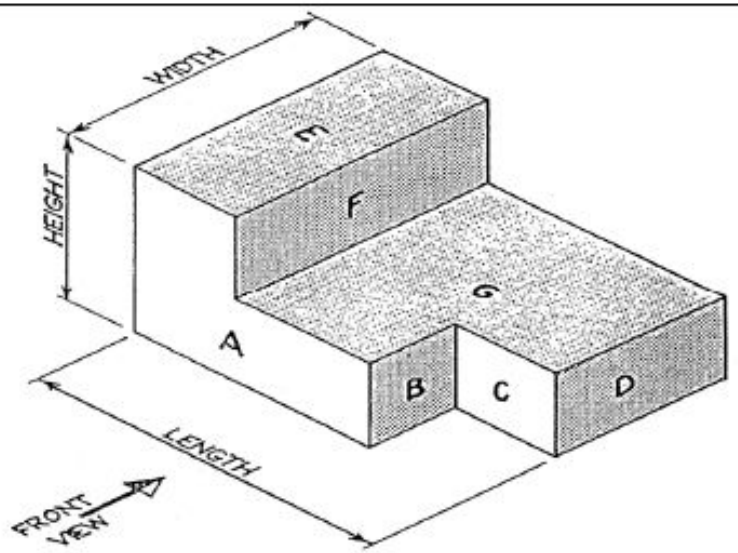
TOP VIEW



FRONT VIEW



SIDE VIEW

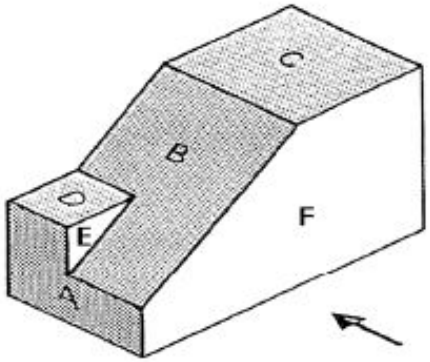


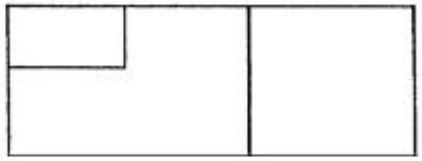
A	B	C	D	E	F	G

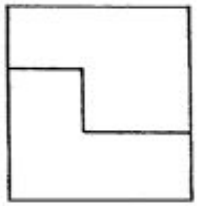
DRG.	ORTHOGRAPHIC PROJECTION	EXERCISE 2

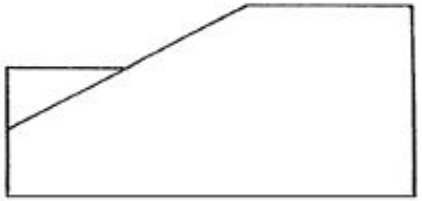
Exercise 1-2-2

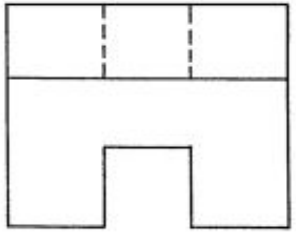
EXERCISES. Transfer the letters from the isometric drawing onto the same plane surfaces of the orthogonal drawing. Name each view.

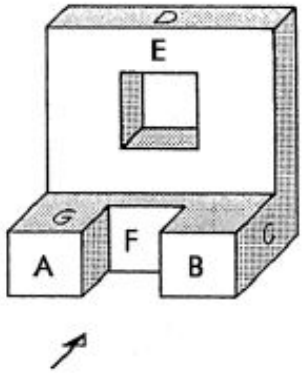


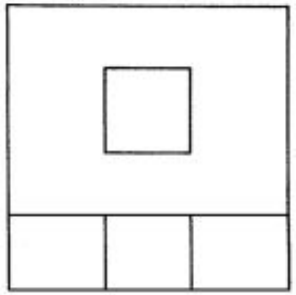


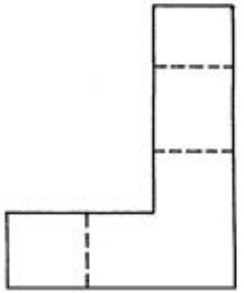












DRG.	ORTHOGRAPHIC PROJECTION	EXERCISE 3

EXERCISES. From drawings 1 to 18 opposite select the view which is requested in the table below. Place the number of this view in the

Orthographic Projection

Activity: *(3 mins)*

Ok… now try this …

- ***Exercise 1-2-3***

Hint: *you need to construct* ***three*** *orthographic views –* ***Front, Plan and Side views*** *– these views are* ***'projected'*** *off each*

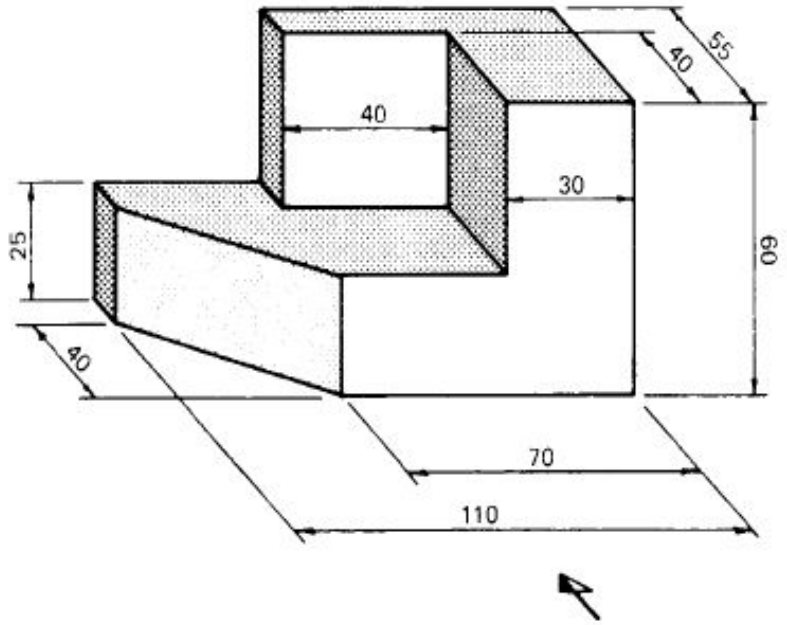
Start with the ***'Front View'*** *– look in the direction of the* ***arrow***

You can use ***'squared' paper*** *if you prefer*

Orthographic Projection

Exercise 1-2-3 – *3mins*

EXERCISES. **FREEHAND SKETCHING.** Sketch, on the space provided, the third angle projection of the component below.

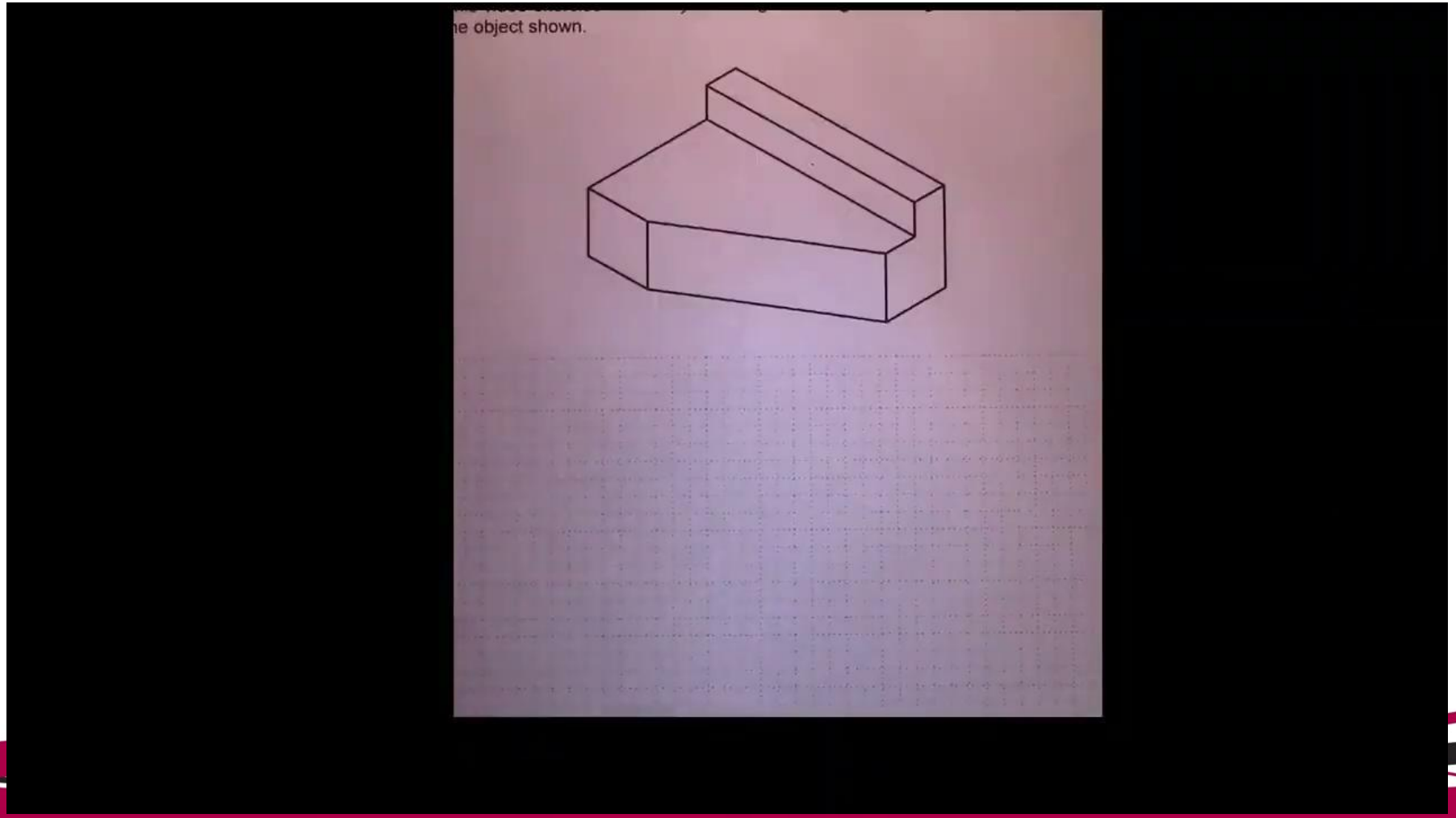


DRG.	ORTHOGRAPHIC PROJECTION	EXERCISE 12

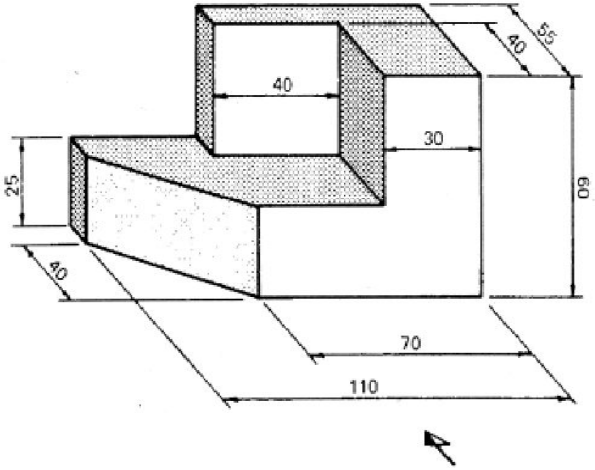
Orthographic Projection

What to draw…

Video 1 - Beginners



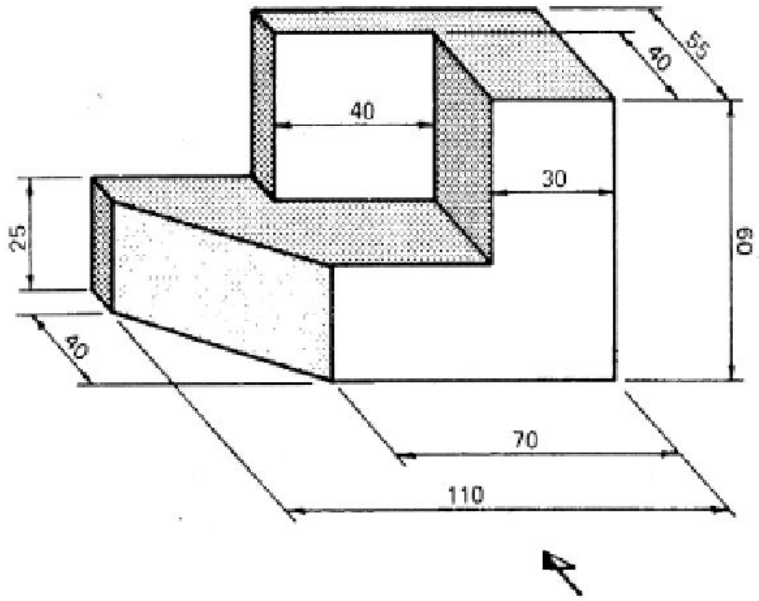
Orthographic Projection

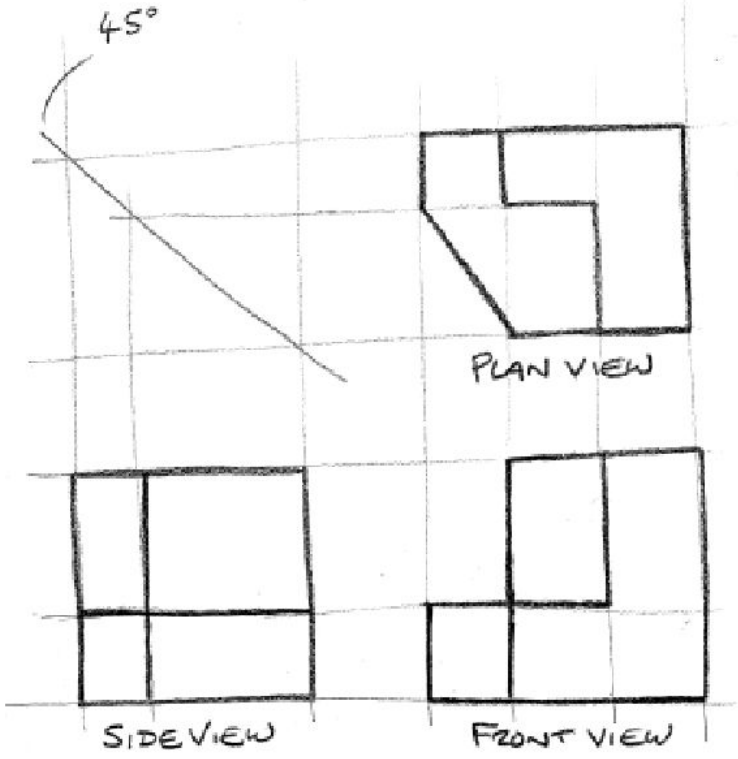


Orthographic Projection

Exercise 1-2-3 Solution - did you get it right??? *You have **2mins** to check and correct your solution*

EXERCISES. FREEHAND SKETCHING. Sketch, on the space provided, the third angle projection of the component below.

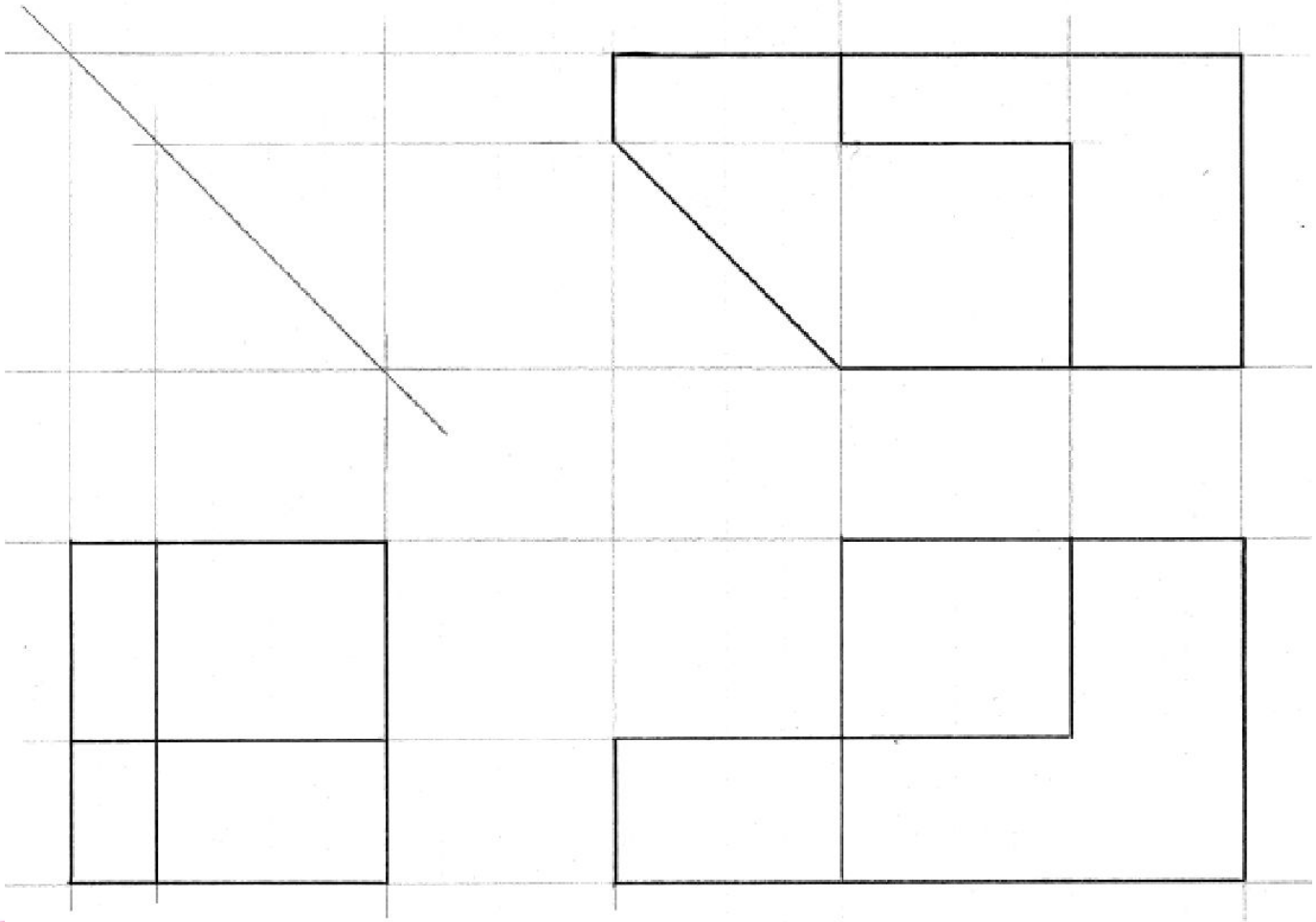




DRG.	ORTHOGRAPHIC PROJECTION	EXERCISE 12

Orthographic Projection

Exercise 1-2-3 Solution - did you get it right??? *You have **2mins** to check and correct your solution*



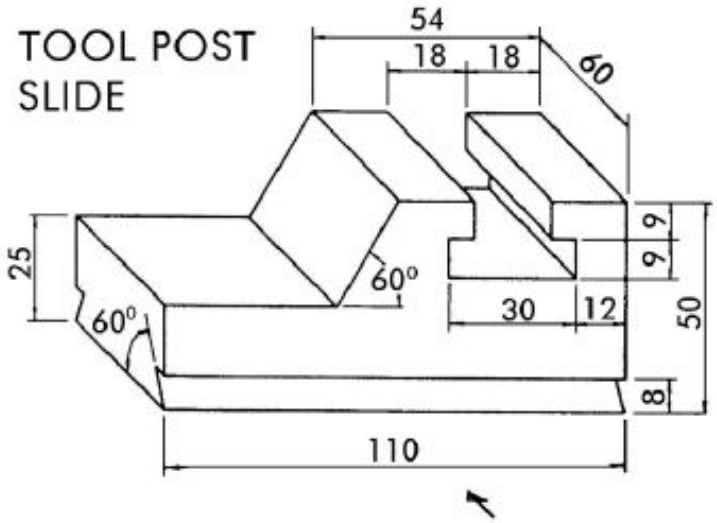
Extension Exercise 1-2-4

- Using the worksheets provided draw the given shapes in third angle orthographic projection.
- Your solution should be drawn on ***'1 cm squared paper'***.

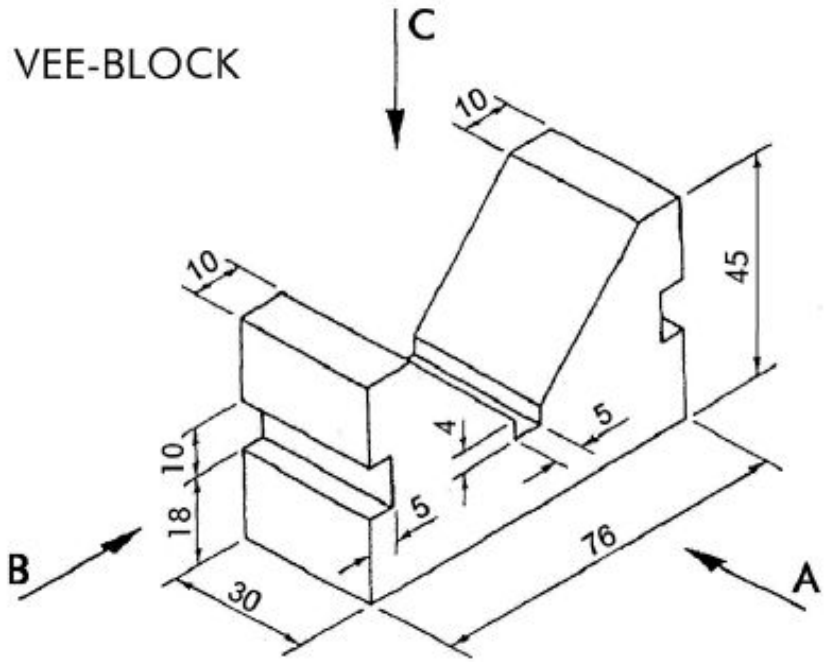
Extension Exercise 1-2-4

Exercises. Draw three views of the following components, either in first or third angle projection. Show hidden detail where necessary and fully dimensions your drawing.

TOOL POST SLIDE



VEE-BLOCK



DRG.	ORTHOGRAPHIC PROJECTION	ALL DIMENSION IN MILLIMETRES

Plenary

Reflective Discussion…

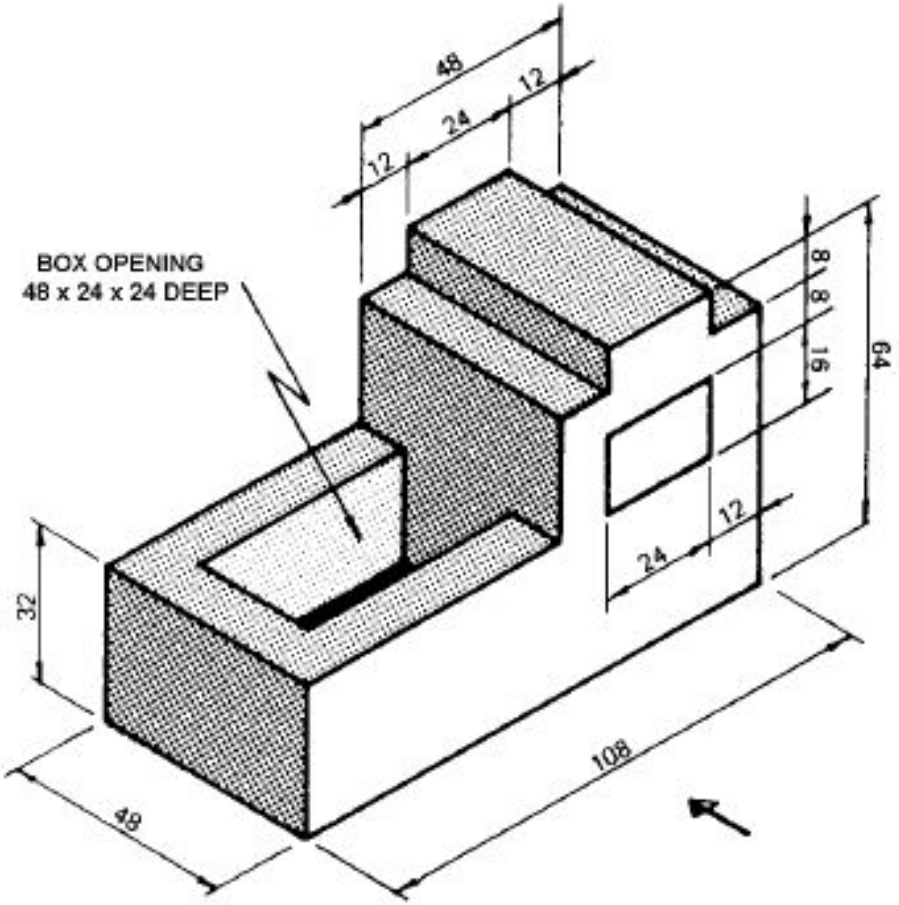
What have you learnt???

- *What is an orthographic drawing?*
- *How many **views** does an orthographic drawing typically consist off?*
- *What are these **views** called?*

Presentation

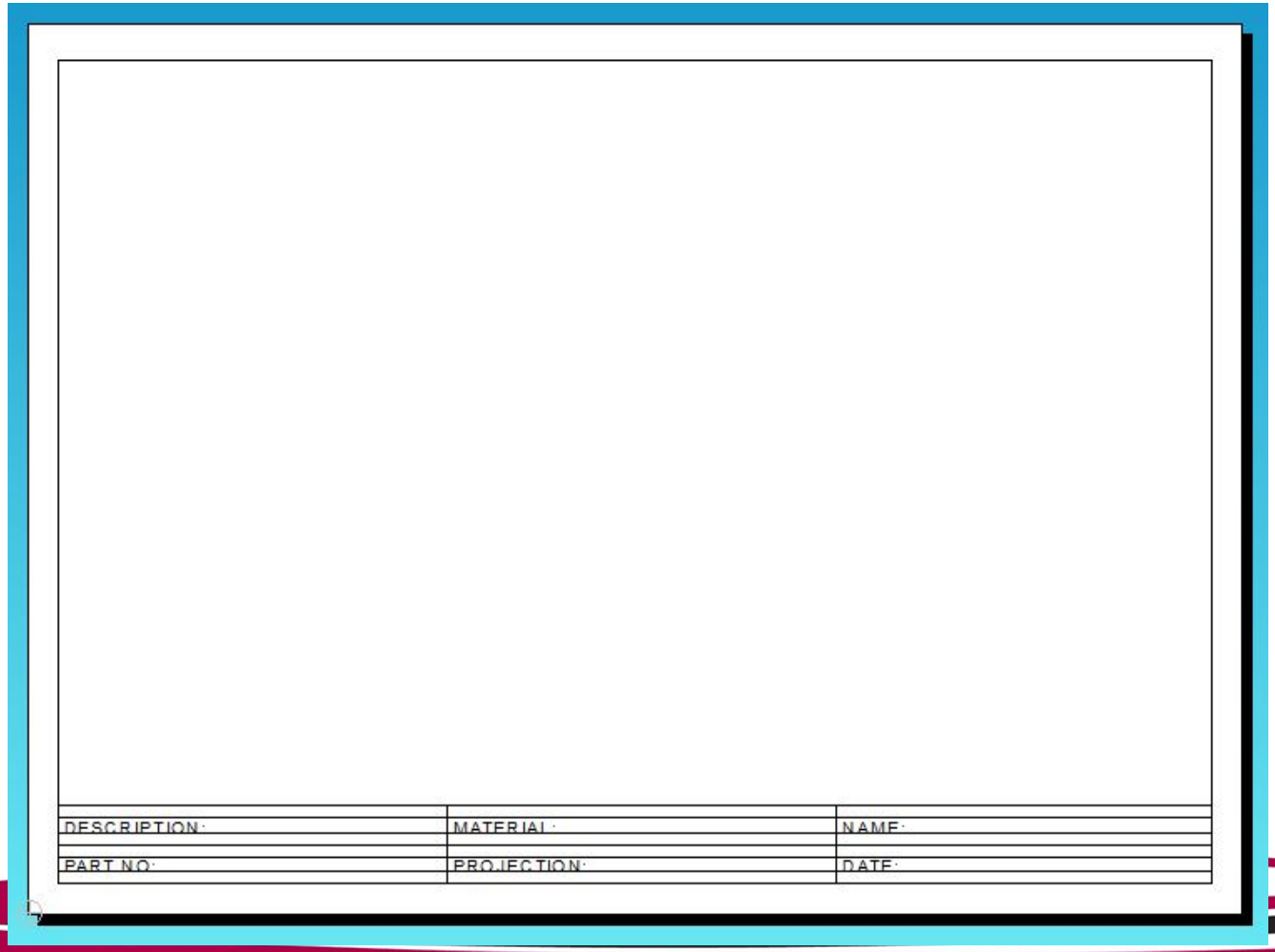
Starter

Using 3rd Angle Projection and 5mm squared paper – produce an orthographic drawing of the object below:



Presentation

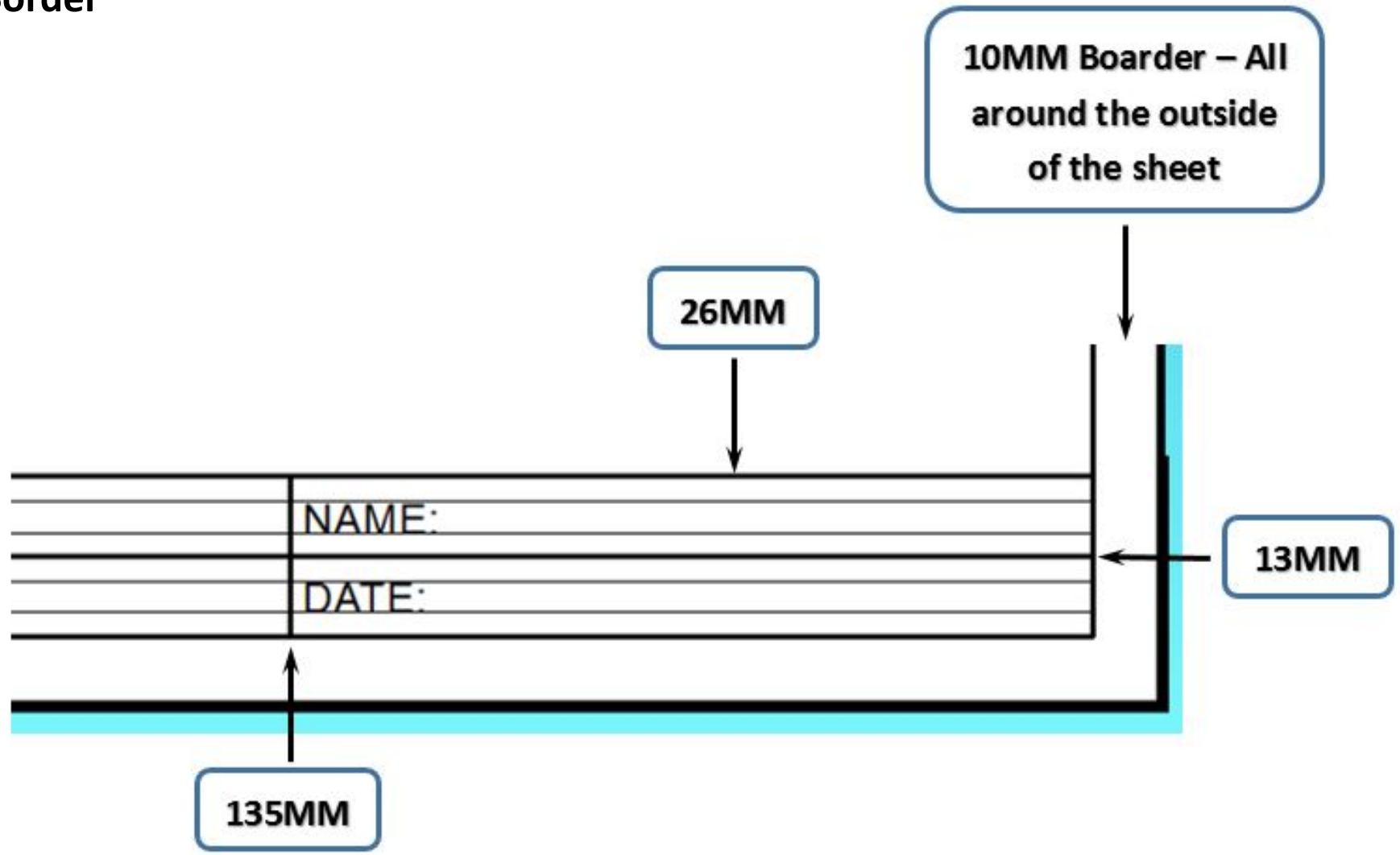
When drawing formally we use drawing boards, A3 paper and set squares. All constructions lines can be left in, however they must be drawn in feint pencil.



Your work must be presented neatly on A3 paper. You must also add a border and title bar (see opposite)

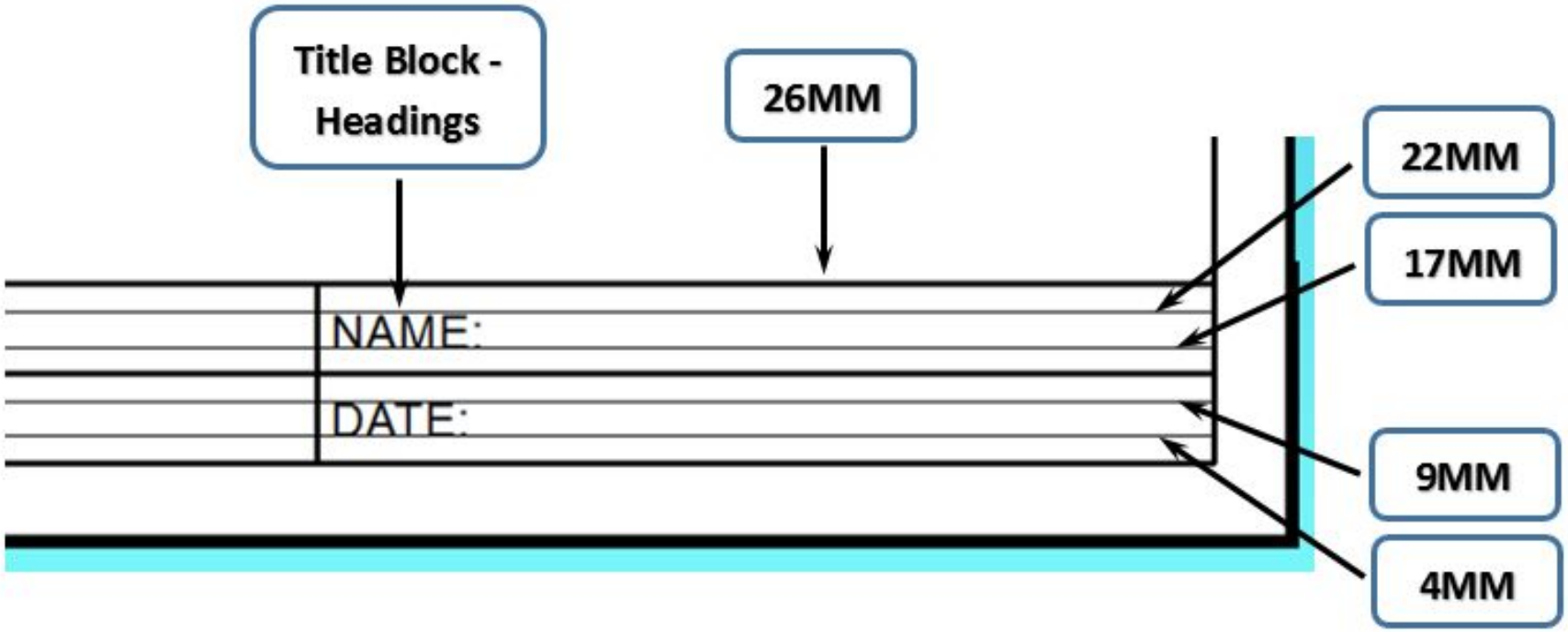
Presentation

Border

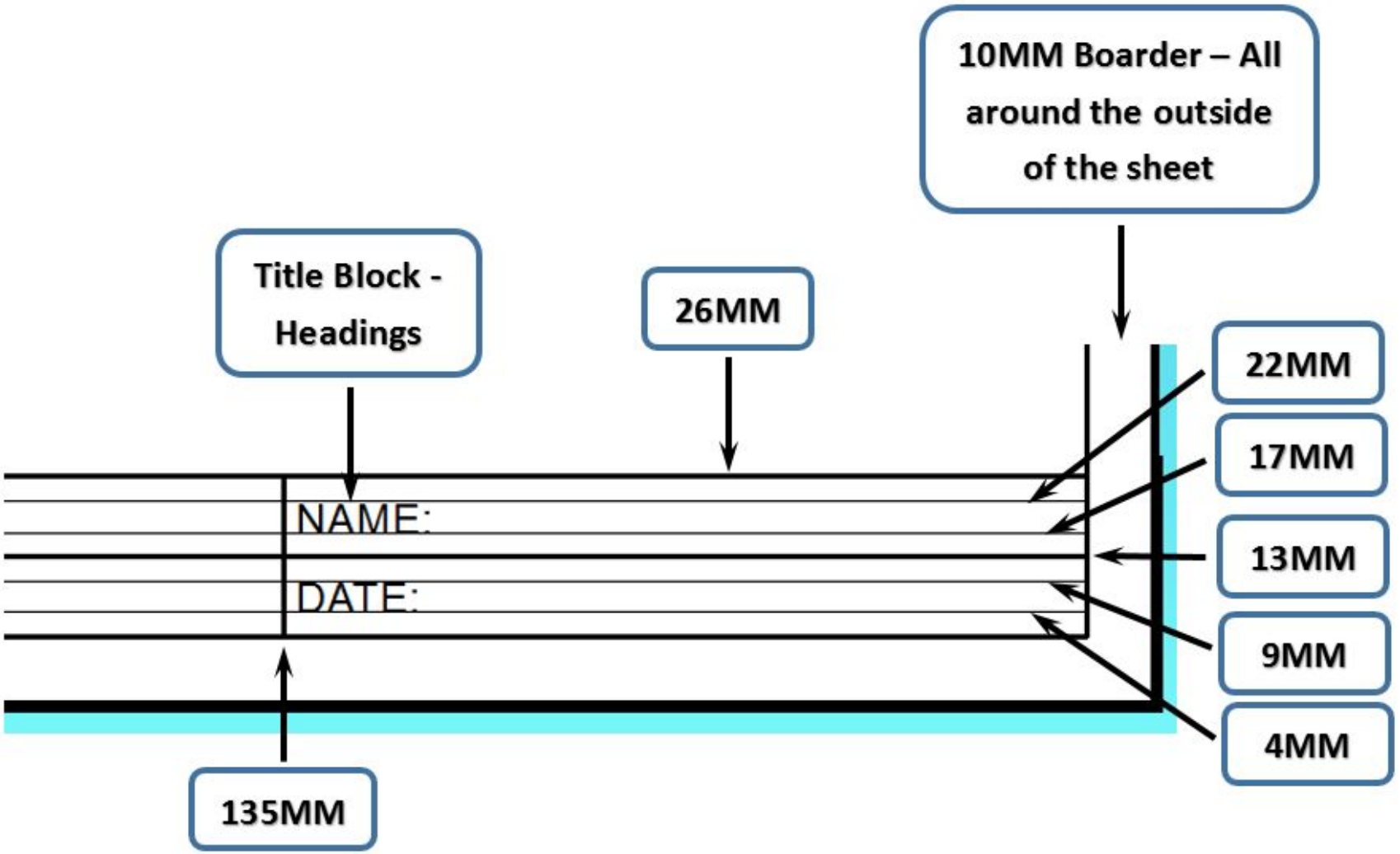


Presentation

Guide Lines and Title Printing



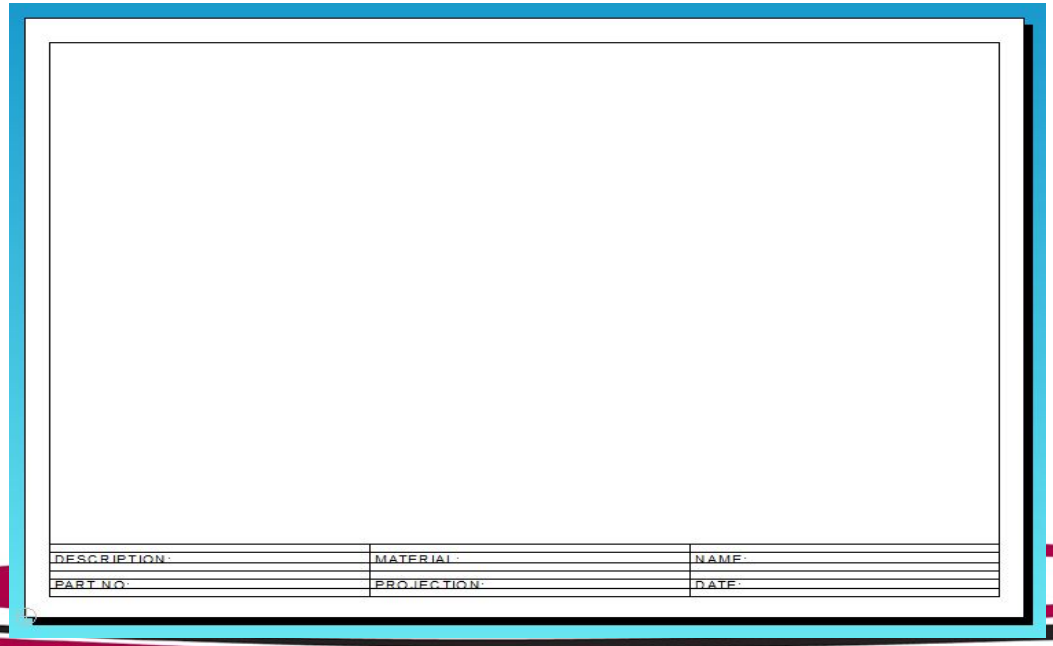
Presentation



Presentation

Headings for your Title Block

DESCRIPTION:	MATERIAL:	NAME:
PART NO:	PROJECTION:	DATE:



Your work must be presented neatly on A3 paper. You must also add a border and title bar (see opposite)

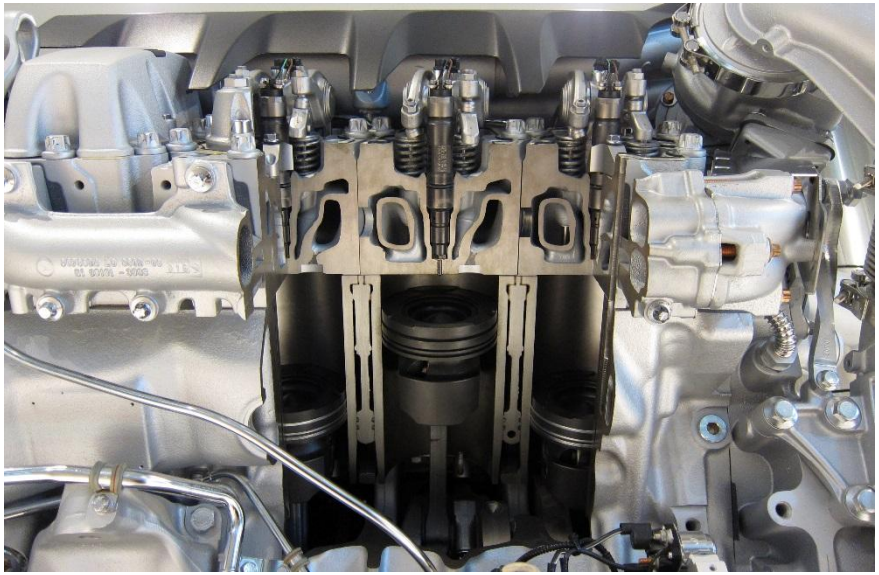
Creating Section Views

Orthographic Projection
Starter 1-4

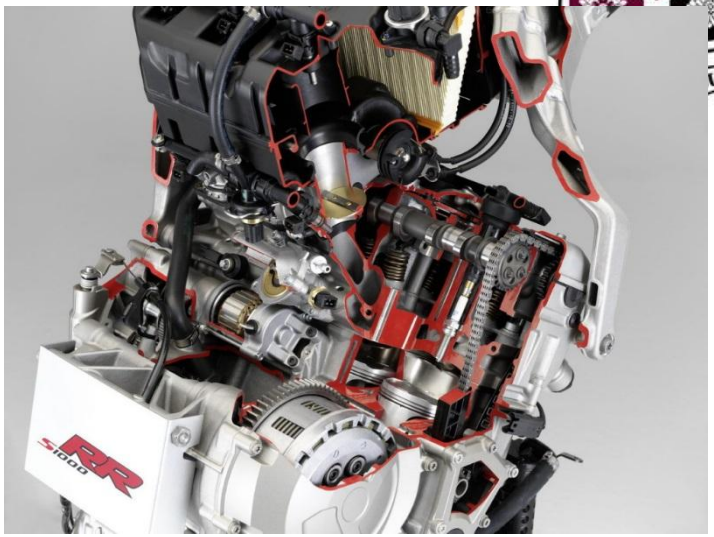
Discuss in pairs – 2mins

- *What do you think a 'Section View' is???*
- *Why might they be useful???*

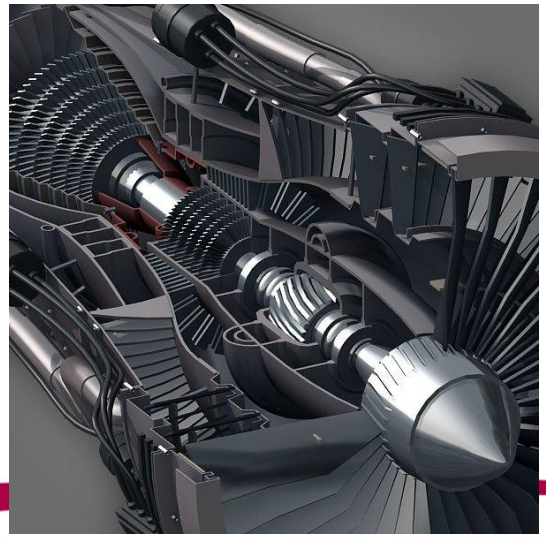
Cut Away Views



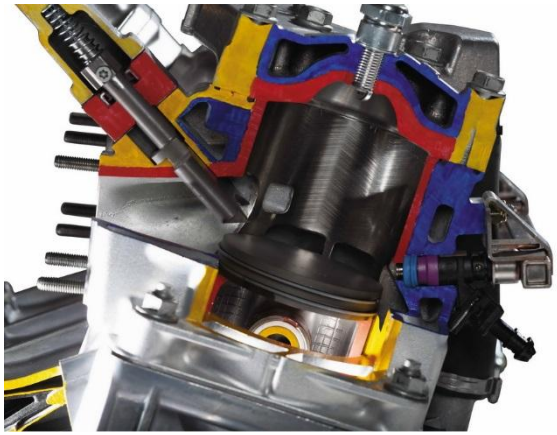
Car Engine



Motor Bike Engine



Jet Engine



Snow Mobile Engine

Learning Objectives

- To understand the need for Section Views on Engineering Drawings
- To be able to construct a Section View on a part in one plane to British Standard *BS8888* using drawing equipment - drawing boards, rules, set square, compasses etc.

Learning Outcomes – L7,8

- **ALL** students to *know* what a *Section View* is and why they are sometimes necessary on an Engineering Drawing.
- **MOST** students to *understand* how to *construct* a *Section View* in on plane on a *simple part*.
- **SOME** students to be able to *apply* their *understanding* of sections and be able to *construct* a Section View on a *complex part or assembly of parts* to BS8888 standards.

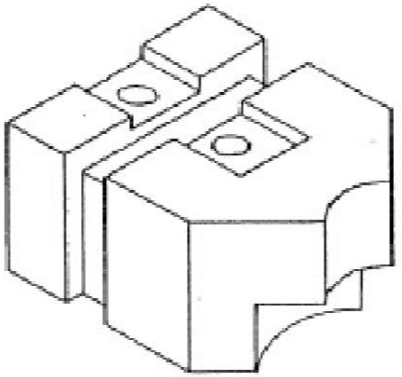
Sectional Views

In many cases the normal views given by orthographic projections are not sufficient to show the complete shape description of an item including the inside detail. It is necessary to draw additional views showing the item cut open to reveal the inside.

These are called ***'sectional views'.***

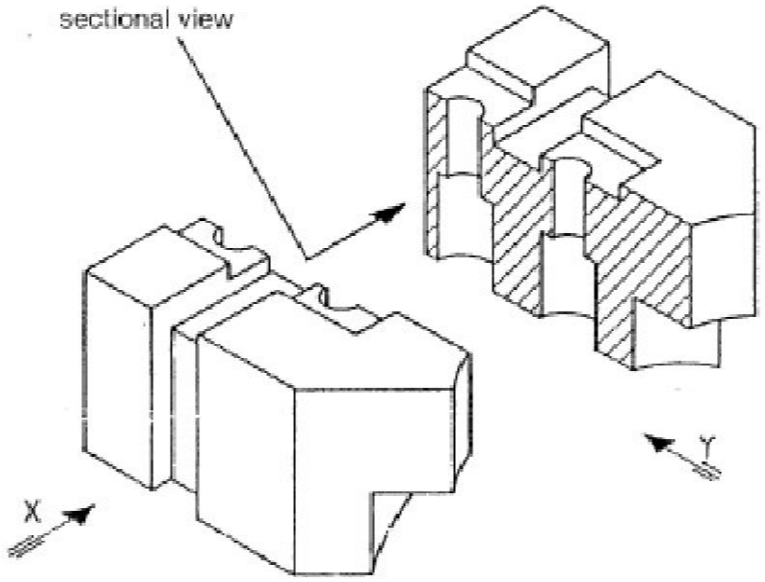
Sectional Views

The isometric view of the component shown has been cut through along its ***centre line*** and the two pieces moved apart.



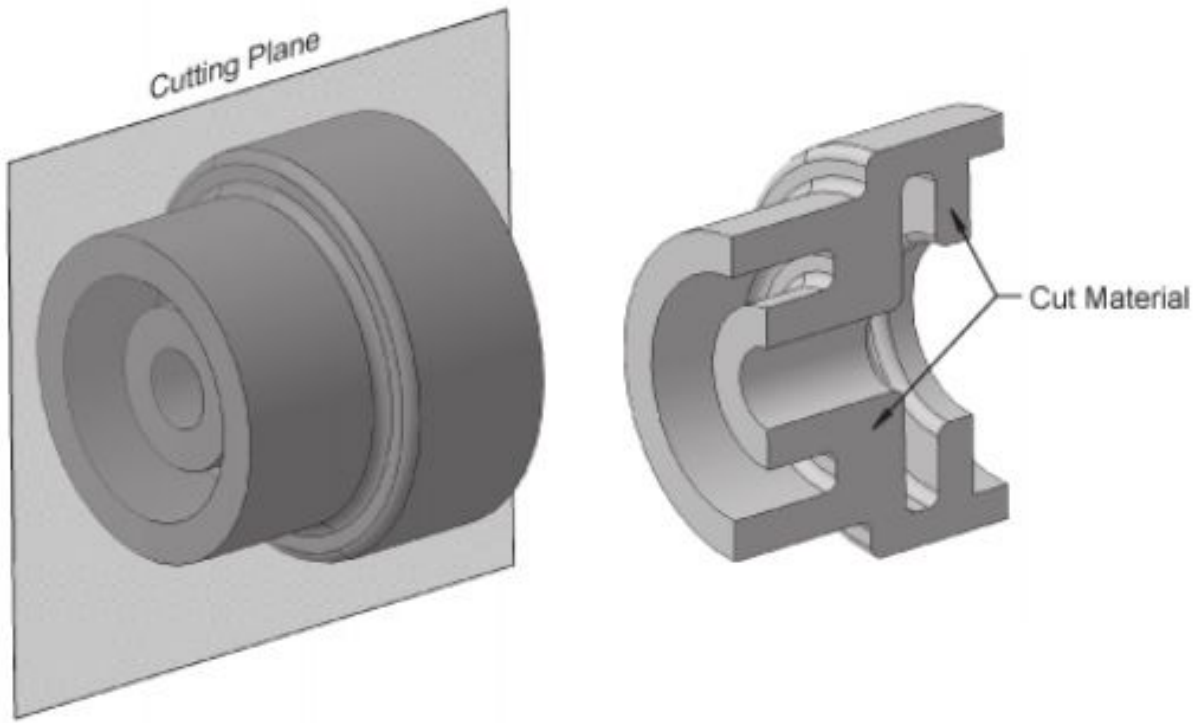
The shape of the ***counter bored*** holes are revealed.

The sectional view is the one with the ***cross-hatched*** face.

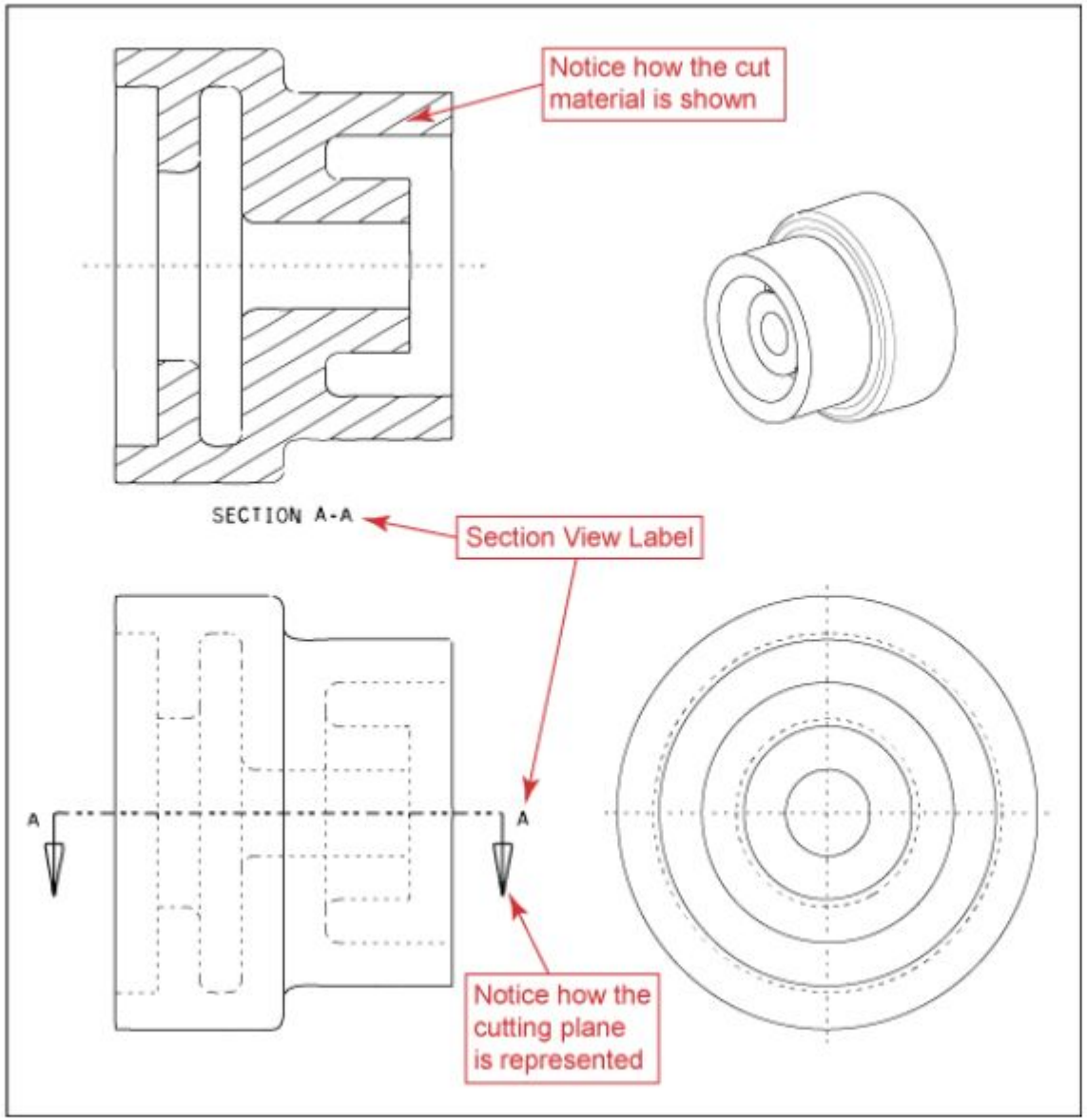


Creating a Sectional View

- The part is cut using an imaginary cutting plane.
- The unwanted portion is mentally discarded exposing the interior construction.

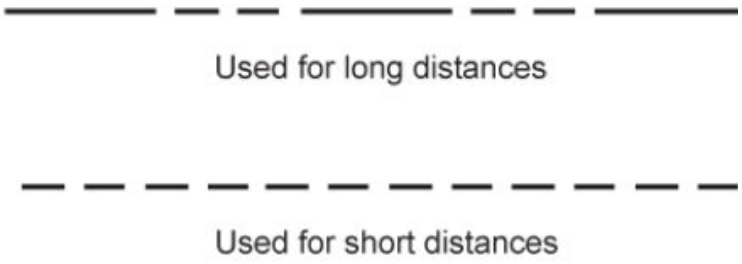


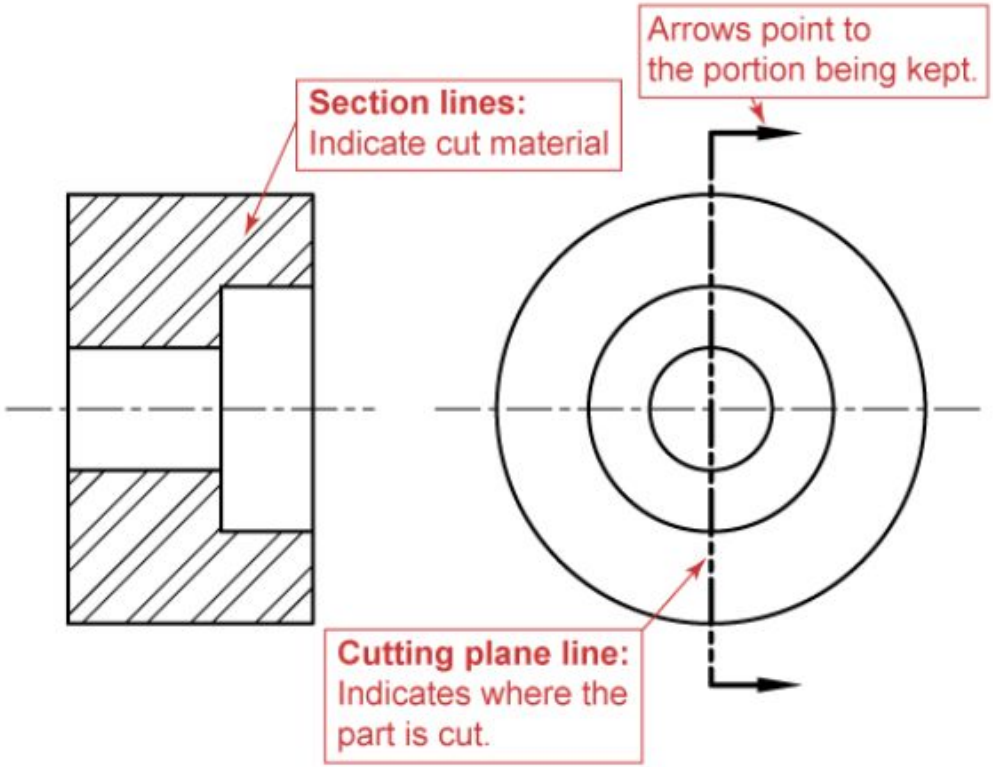
Example of a Sectional View



Lines used in Sectional Views

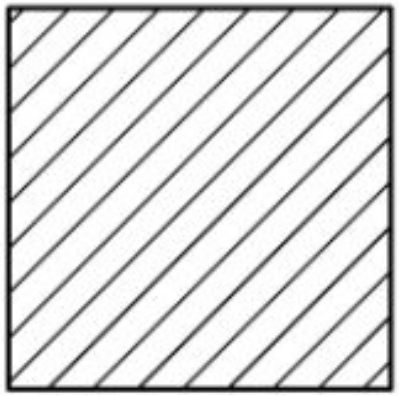
Cutting Plane Line: Used to show where the object is being cut.



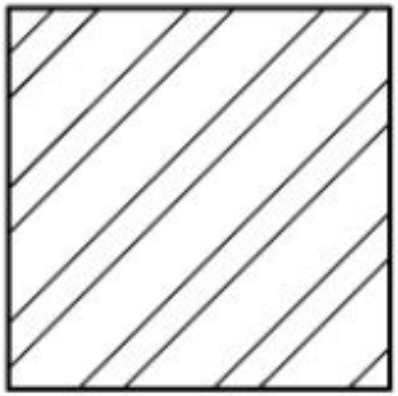


Section Lines

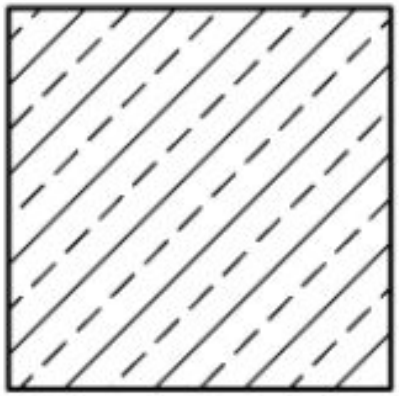
Used to indicate where the cutting plane cuts the material. Section lines are thin and the symbols (type of lines) are chosen according to the material of the object. Section lines are generally drawn at a 45° angle.



Cast Iron,
General use
all materials



Steel



Brass, Bronze,
Copper

Rules for Sectioning

Rule 1: A section lined area is always completely bounded by a visible outline.

Rule 2: The section lines in all areas should be parallel. Section lines shown in opposite directions indicate a different part.

Rule 3: All the visible edges behind the cutting plane should be shown.

Rule 4: Hidden features should be omitted in all areas of a section view. Exceptions include threads and broken out sections.

What type of section should I use?

The type of section used depends on the situation and what information needs to be conveyed.

Basic Sections

- *Full Section*
- *Half Section*
- *Offset Section*

Section Views in One Plane

See videos below:

Video 1

https://www.youtube.com/watch?v=uL14DZlitqI

Video 2

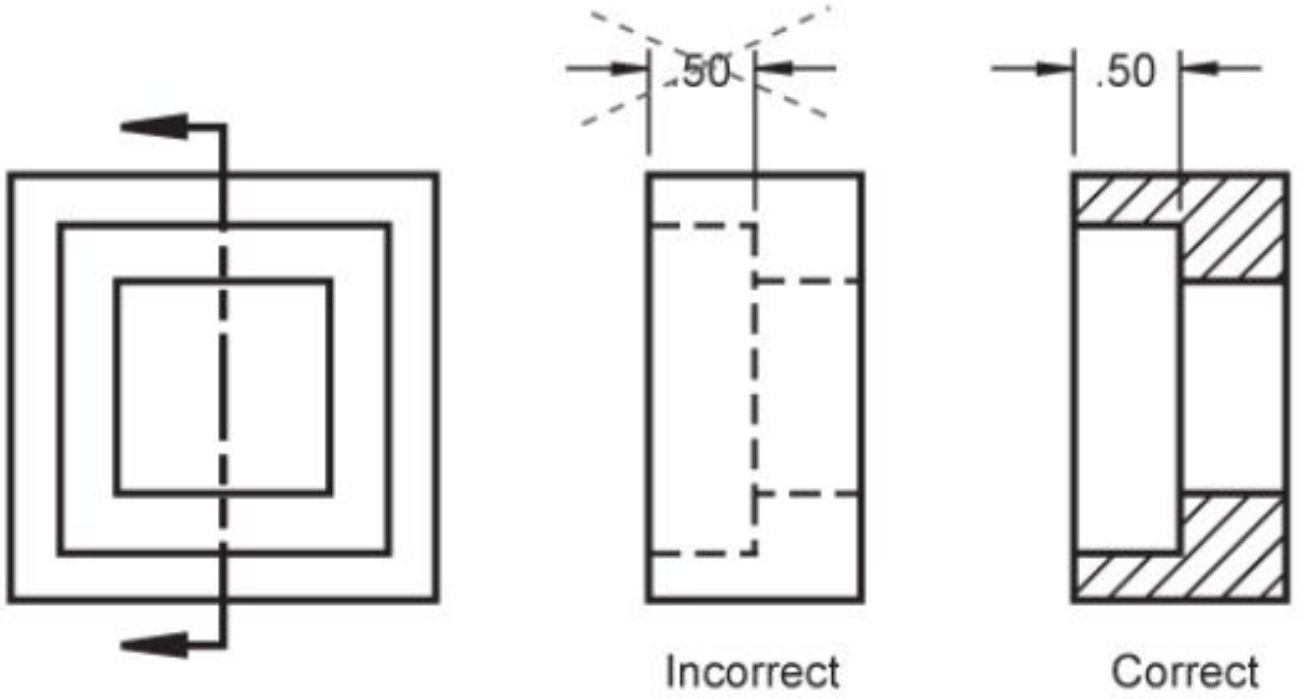
https://www.youtube.com/watch?v=Am0u4H2rGUs

Video: Sectional views and its types



Full Section

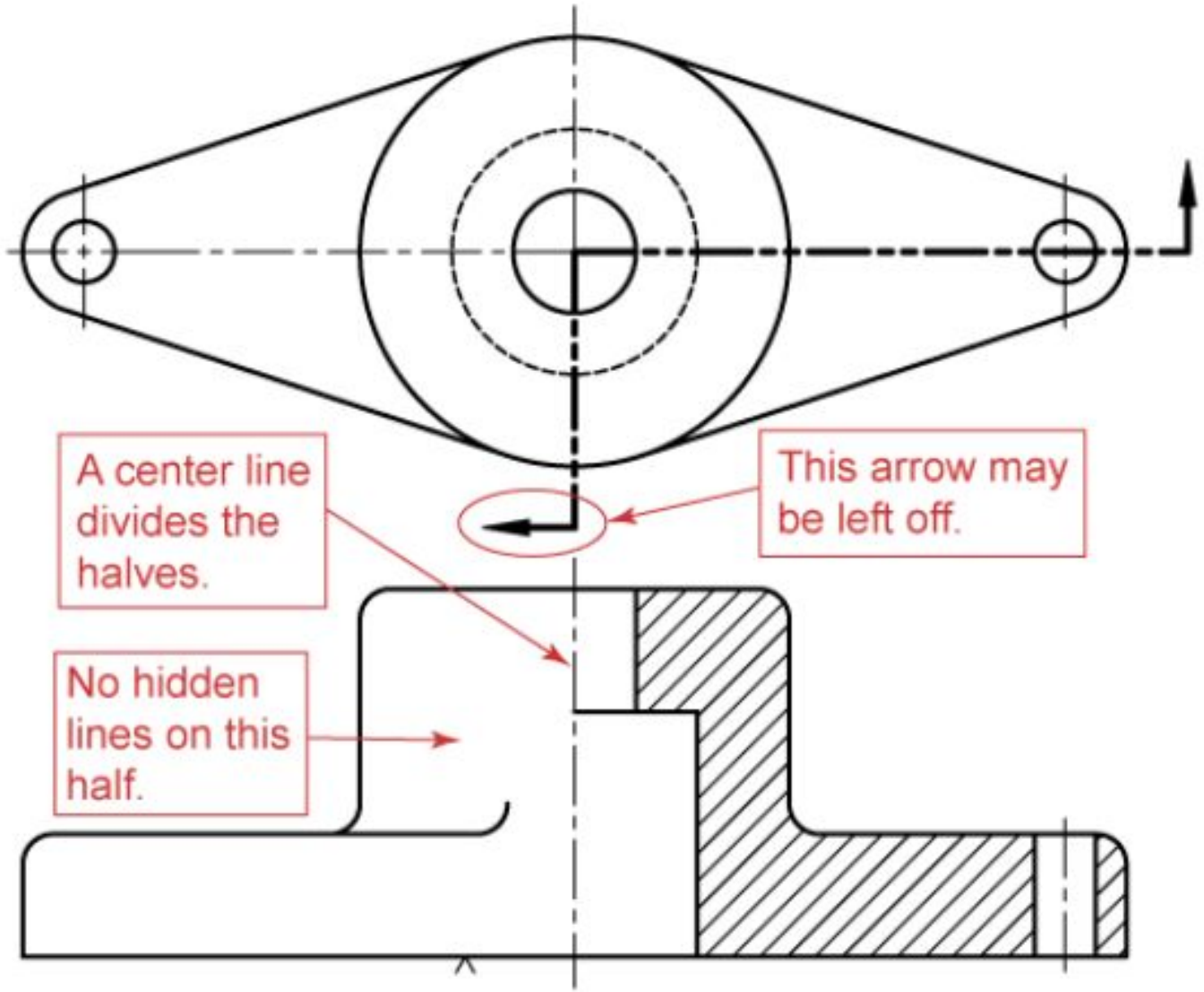
To create a full section, the cutting plane passes fully through the object. Used in many cases to avoid having to dimension hidden lines.



Half Section

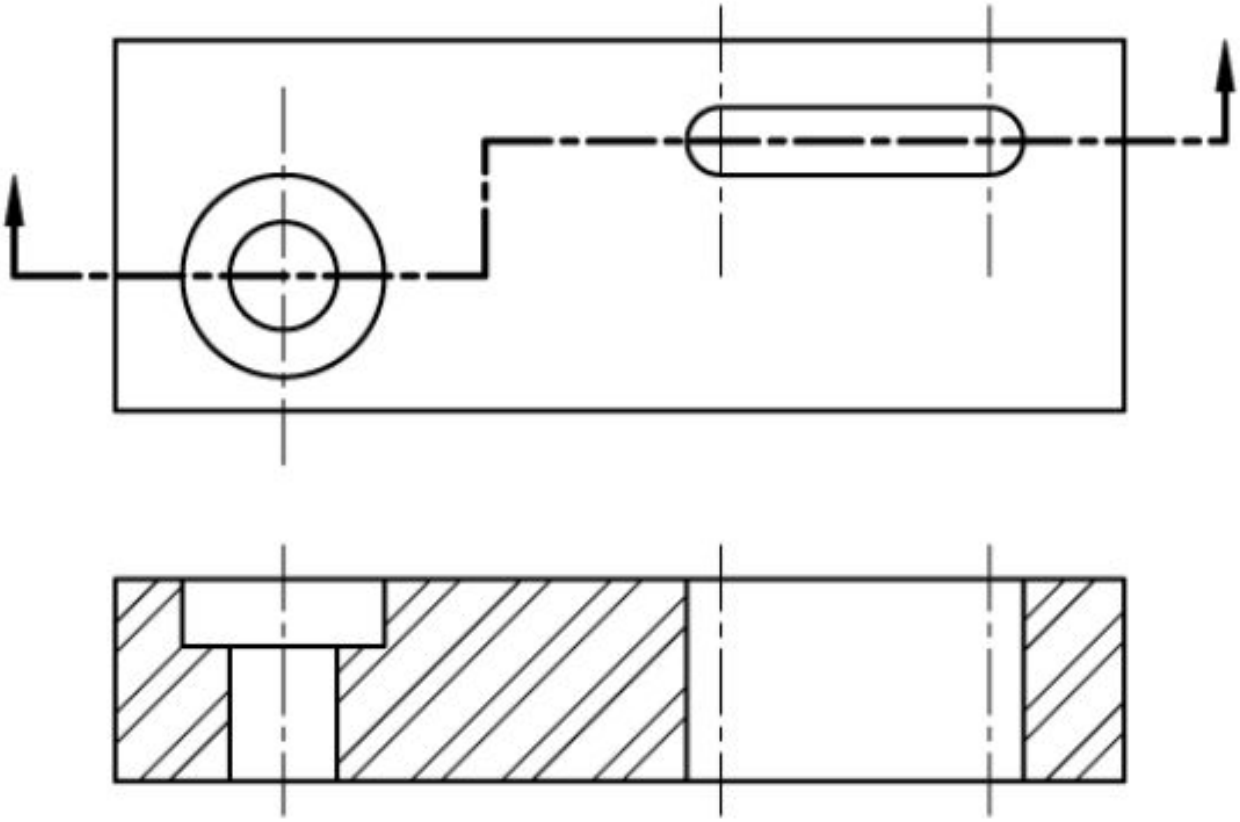
A half section exposes the interior of one half of an object while retaining the exterior of the other half. Half sections are used mainly for symmetric objects or assembly drawings. A centreline is used to separate the two halves. Hidden lines should not be shown on either half.

Half Section



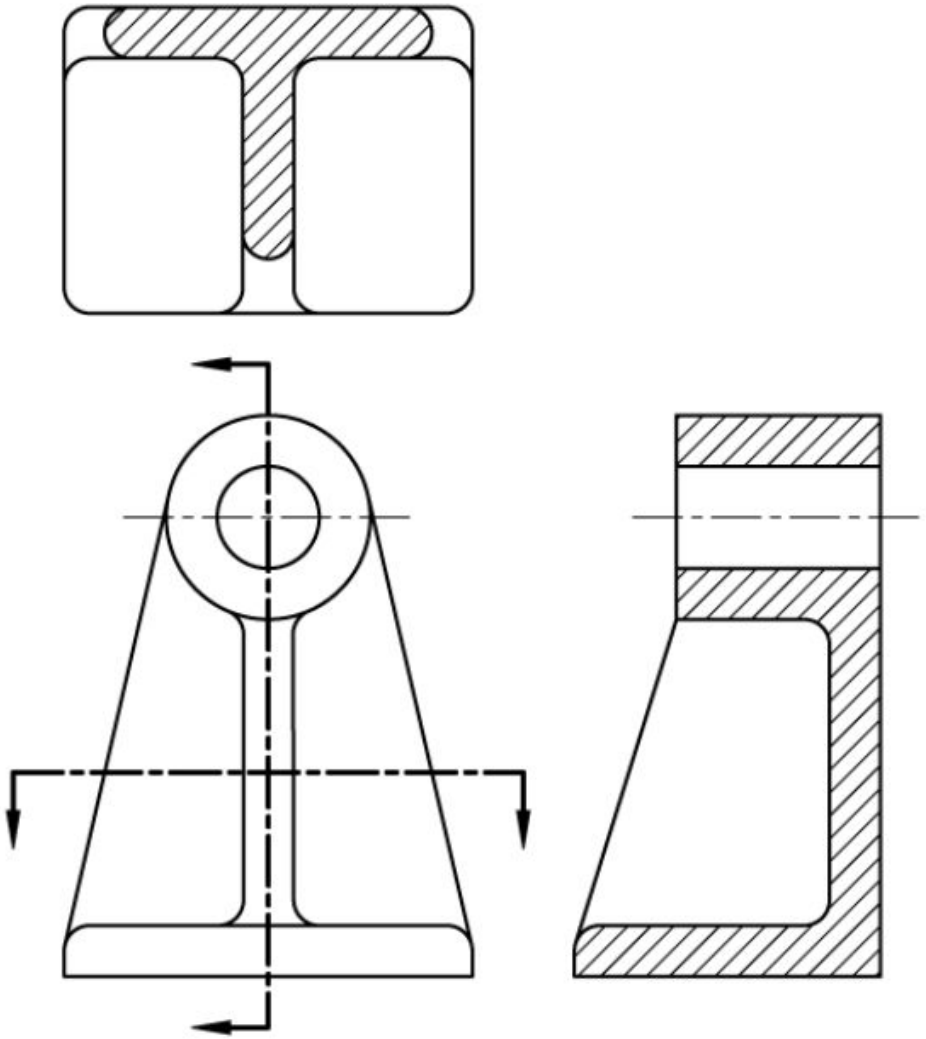
Offset Section

An offset section is produced by bending the cutting plane to show features that don’t lie in the same plane.



Rib and Web Sections

To avoid a false impression of thickness and solidity, ribs and webs are ***not*** sectioned. If the cutting plane passes crosswise through the rib or web, we include section lines.



Non-Sectioned Parts

It is common practice to show standard parts 'in the round' or un-sectioned.

- *Nuts*
- *Bolts*
- *Rivets*
- *Shafts*
- *Screws*

Section Views

Exercise 1-4-1:

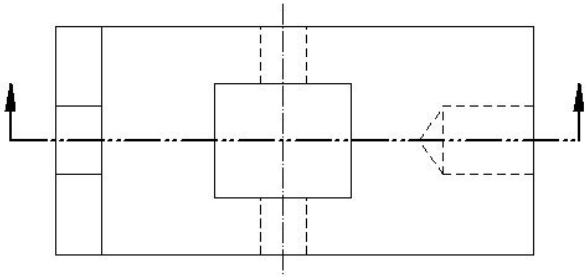
Using A4 paper *'sketch'* the solution to the problem below – You have 10 mins.

Section Views

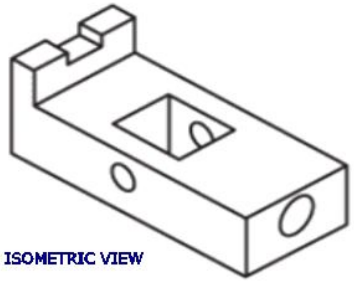
HEATHROW AVIATION ENGINEERING UTC

Exercise 1-4-1

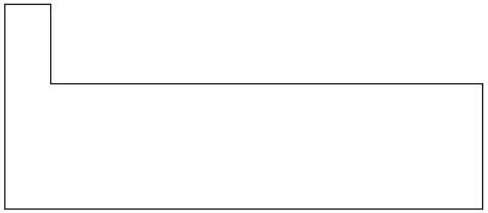
Using the worksheet provided - sketch a ***'full section'*** on the **front view** using the ***'cutting plane'*** *shown* in the **plan view** - You have 10 mins.



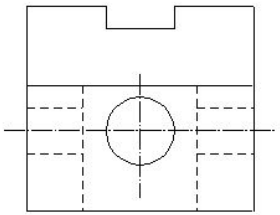
PLAN VIEW



ISOMETRIC VIEW



FRONT VIEW



SIDE VIEW

Section Views

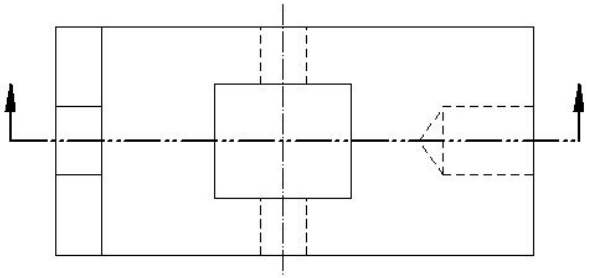
Exercise 1-4-1 - Solution

Section Views

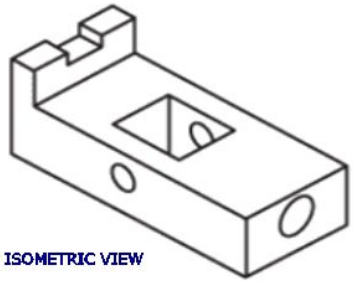
HEATHROW AVIATION ENGINEERING UTC

Exercise 1-4-1 - Solution

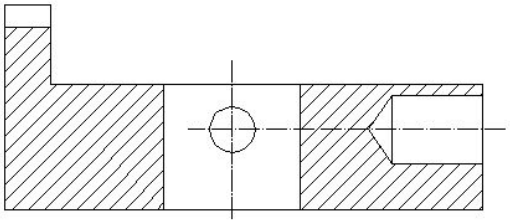
Using the worksheet provided - sketch a ***'full section'*** on the **front view** using the ***'cutting plane'*** *shown* in the **plan view** - You have 10 mins.



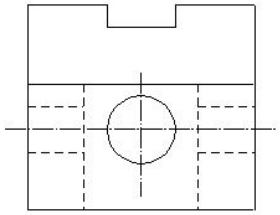
PLAN VIEW



ISOMETRIC VIEW



FRONT VIEW

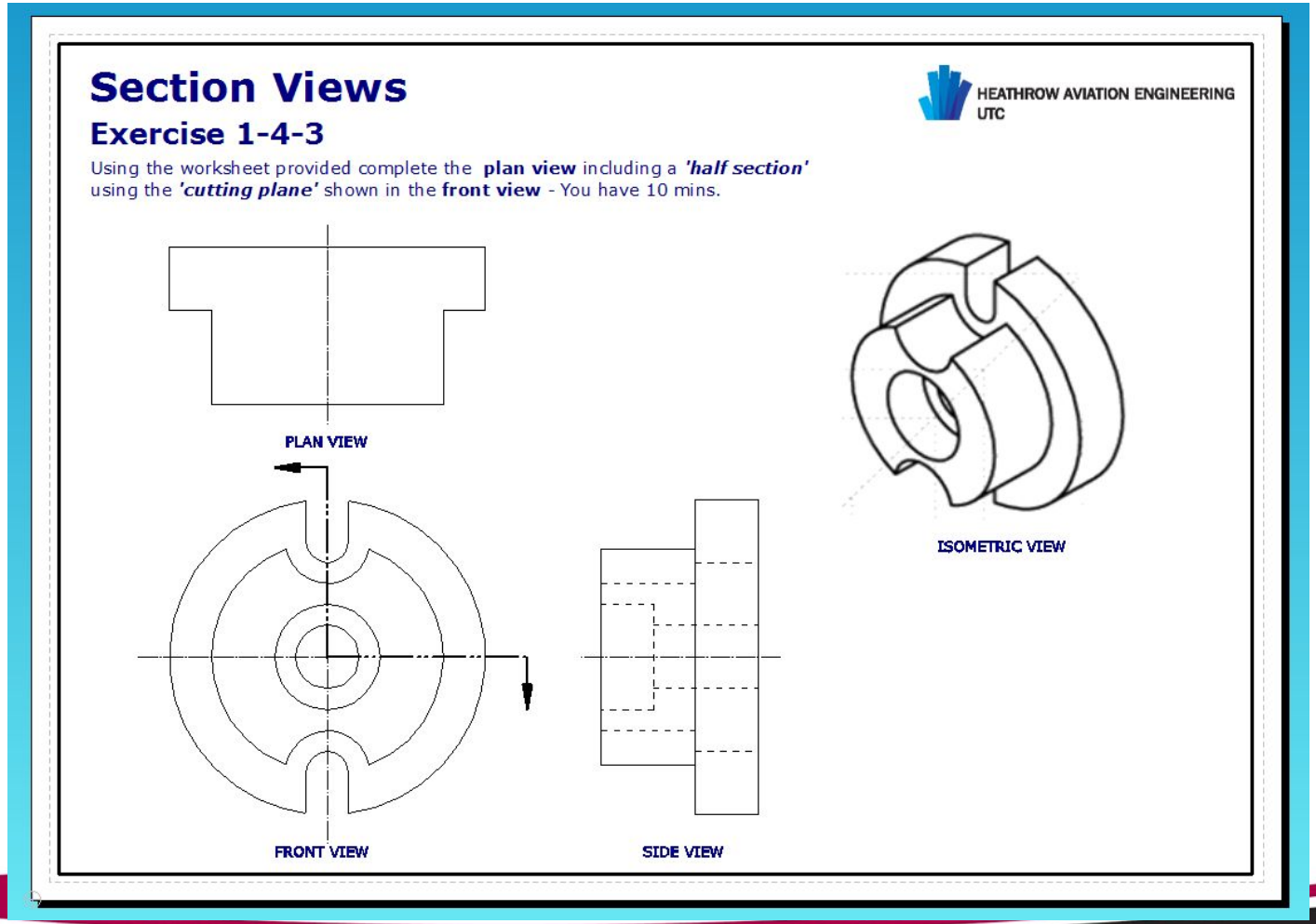


SIDE VIEW

Section Views

Exercise 1-4-3:

Using A4 paper *'sketch'* the solution to the problem below – You have 10 mins.



Section Views

Exercise 1-4-3:

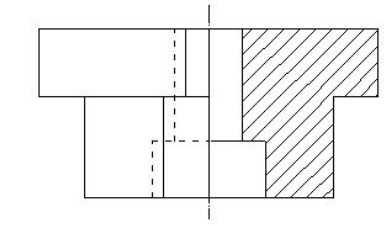
Using A4 paper *'sketch'* the solution to the problem below – You have 10 mins.

Section Views

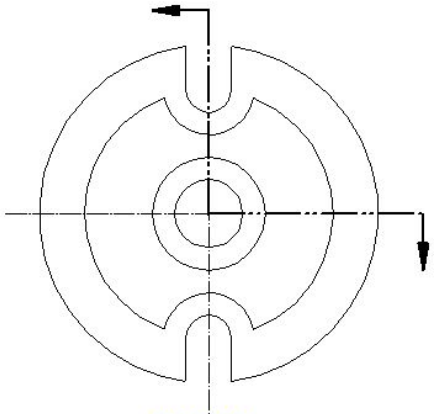
Exercise 1-4-3 - Solution

HEATHROW AVIATION ENGINEERING UTC

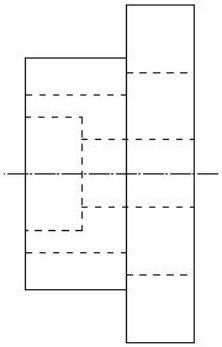
Using the worksheet provided complete the **plan view** including a ***'half section'*** using the ***'cutting plane'*** shown in the **front view** - You have 10 mins.



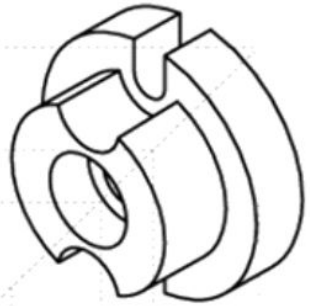
PLAN VIEW



FRONT VIEW



SIDE VIEW



ISOMETRIC VIEW

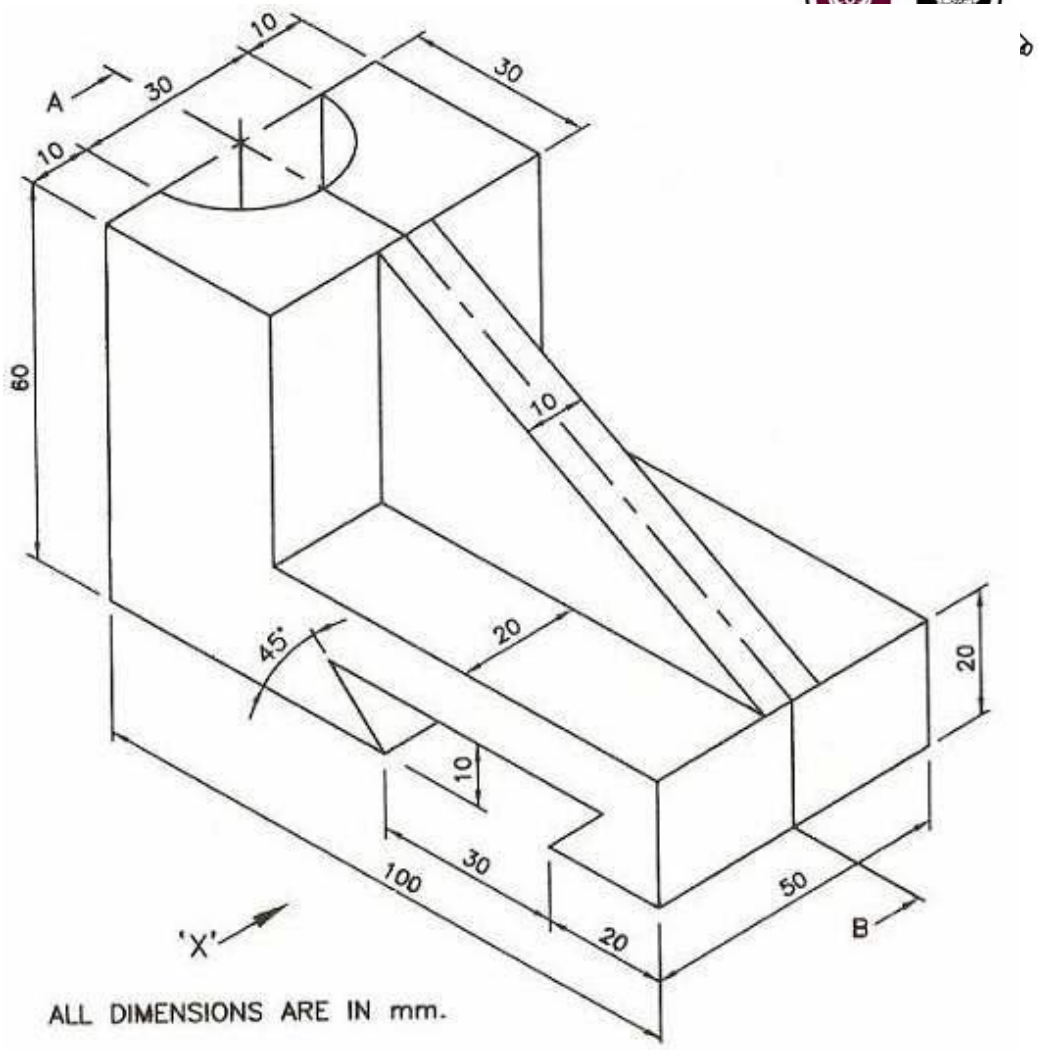
Section Views

Exercise 1-4-2:

Using A3 paper and drawing equipment – add a boarder to the sheet and then construct the solution to the problem below.

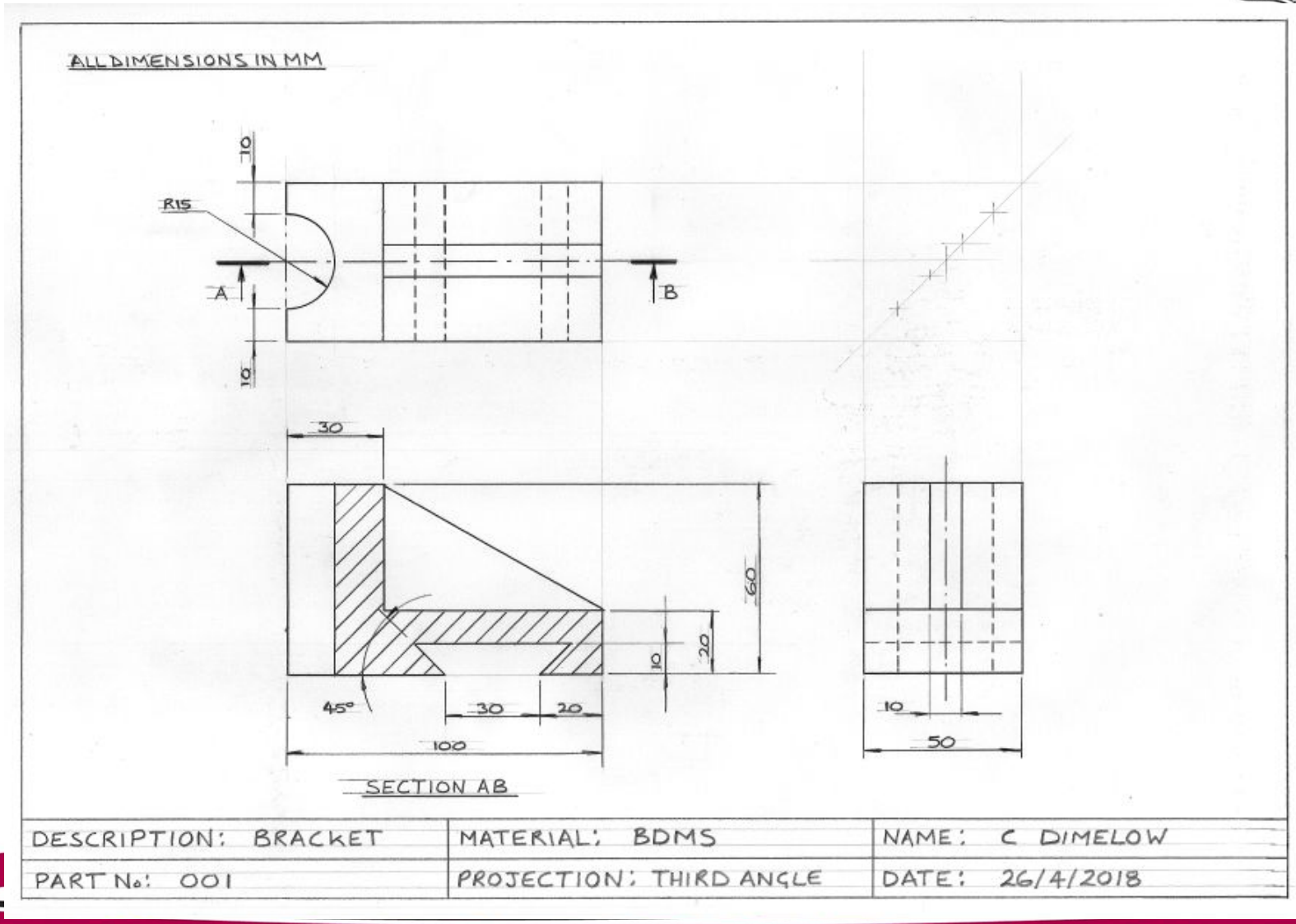
Using the isometric drawing of the machine slide – You have 60 minutes - to construct and fully dimension three orthographic views in *third angle projection*:

a) Construct a ***sectioned*** *front view as indicated* using direction 'X' and the cutting plane A-B
b) Project the *plan view*
c) Project an *end view* to the right of the front view



Section Views

Exercise 1-4-2: Solution



Section Views

Extension 1:

Using A3 paper and drawing equipment – add a boarder to the sheet and then construct the solution to the problem below.

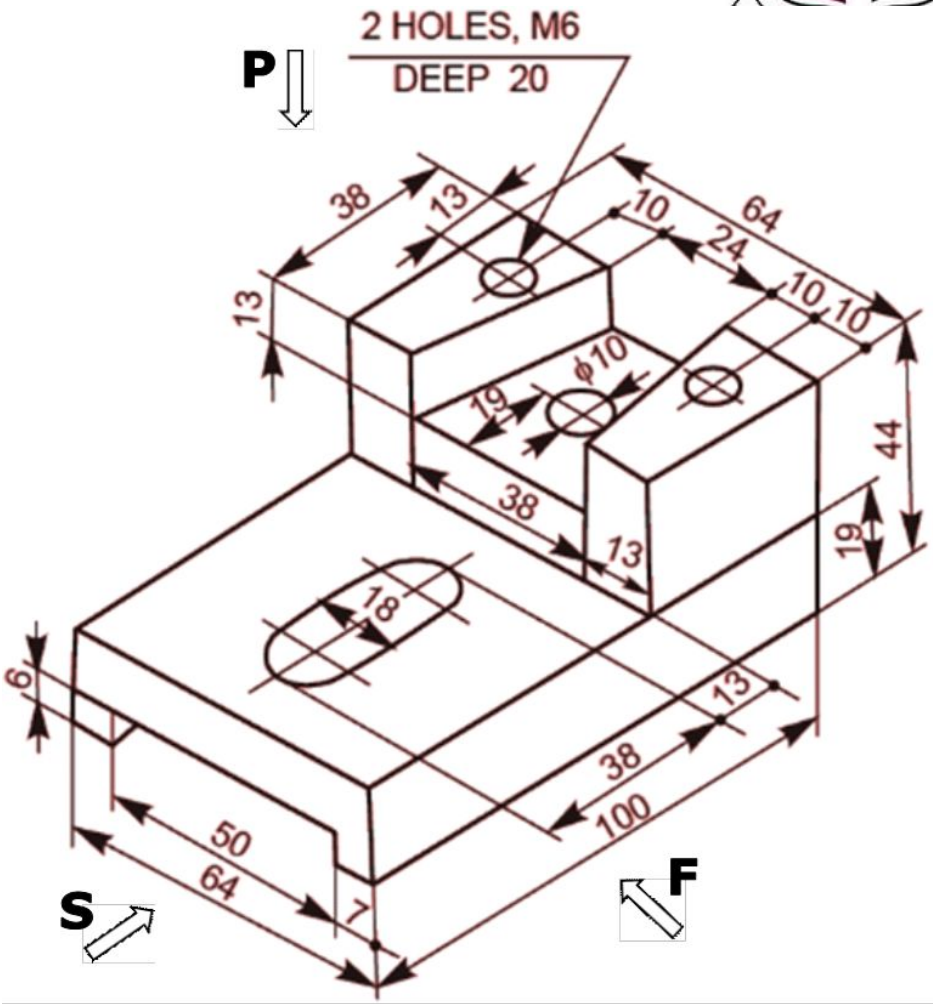
Using the isometric drawing of the machine slide – You have 60 minutes - to construct and fully dimension three orthographic views in *third angle projection*:

1) Construct a front view - direction 'F'

2) Construct a half section side view (direction S) through the bracket on the first hole centre line of the 'slot'.

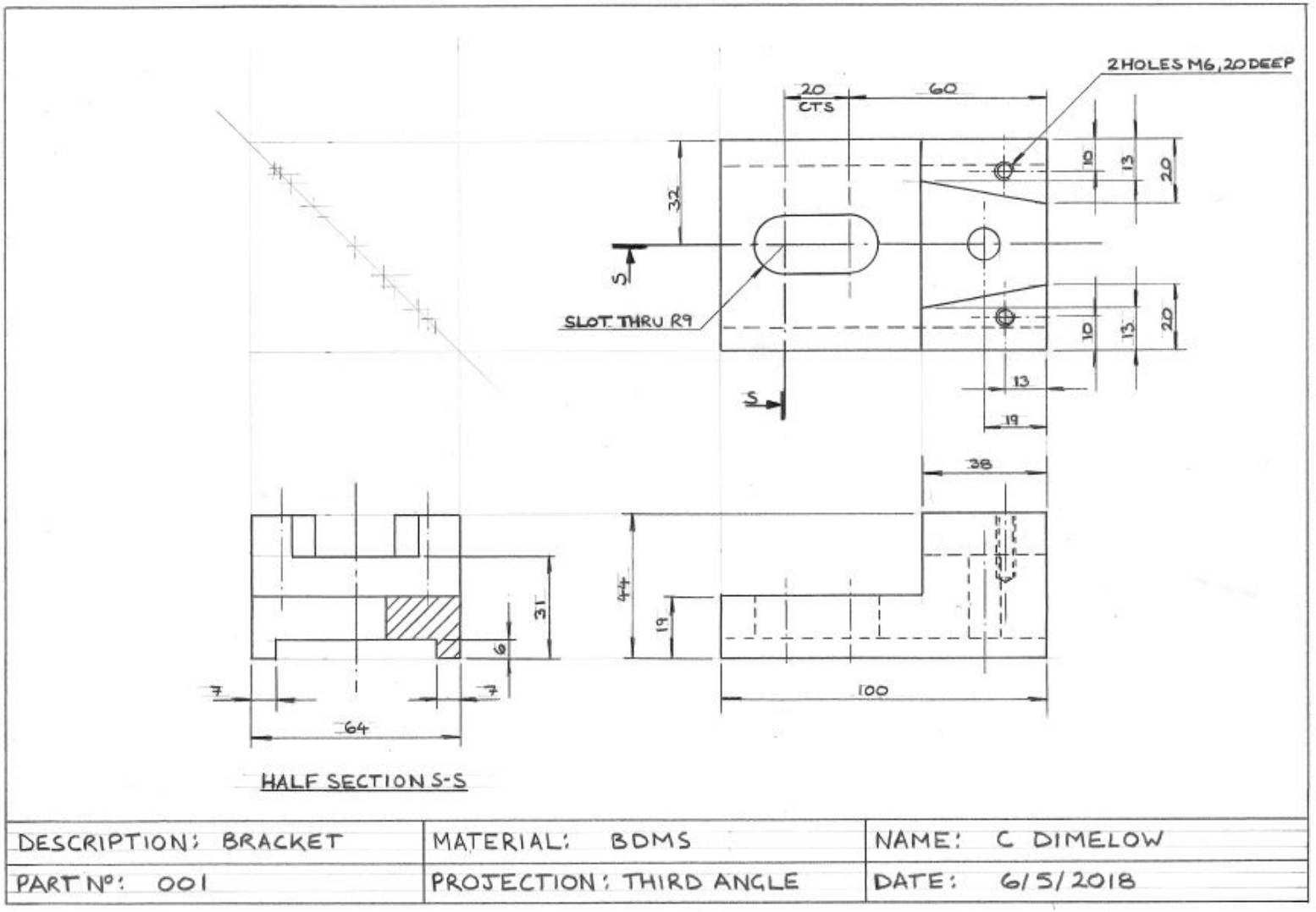
3) Project the plan view - direction 'P'

4) Include all relevant part dimensions and symbols indicating the section.



Section Views

Extension 1: Solution



Section Views

Extension 3:

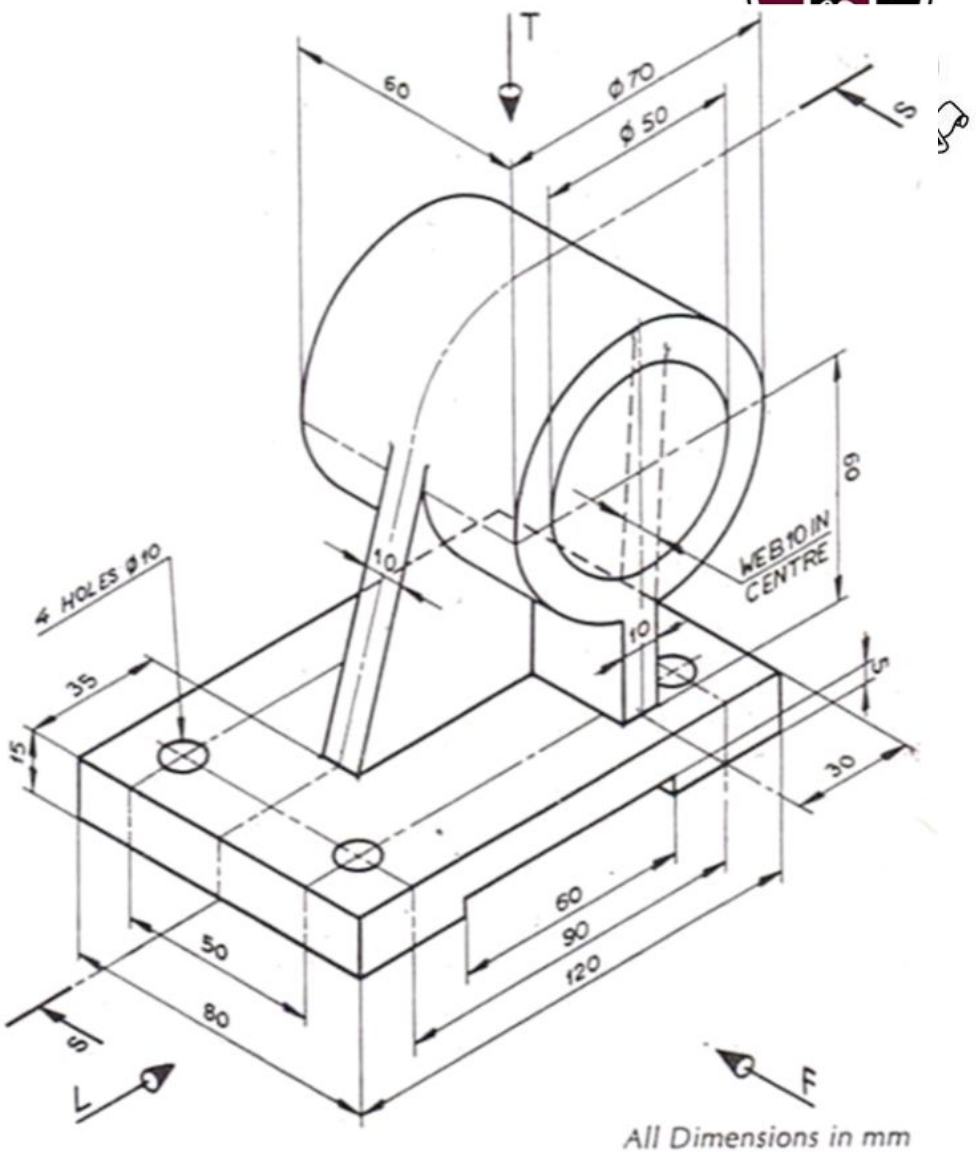
Using A3 paper and drawing equipment – add a boarder to the sheet and then construct the solution to the problem below.

Using the isometric drawing of the machine slide – You have 60 minutes - to construct and fully dimension three orthographic views in *third angle projection*:

1) Construct a sectioned front view as indicated using direction ‘F’ and the cutting plane S-S

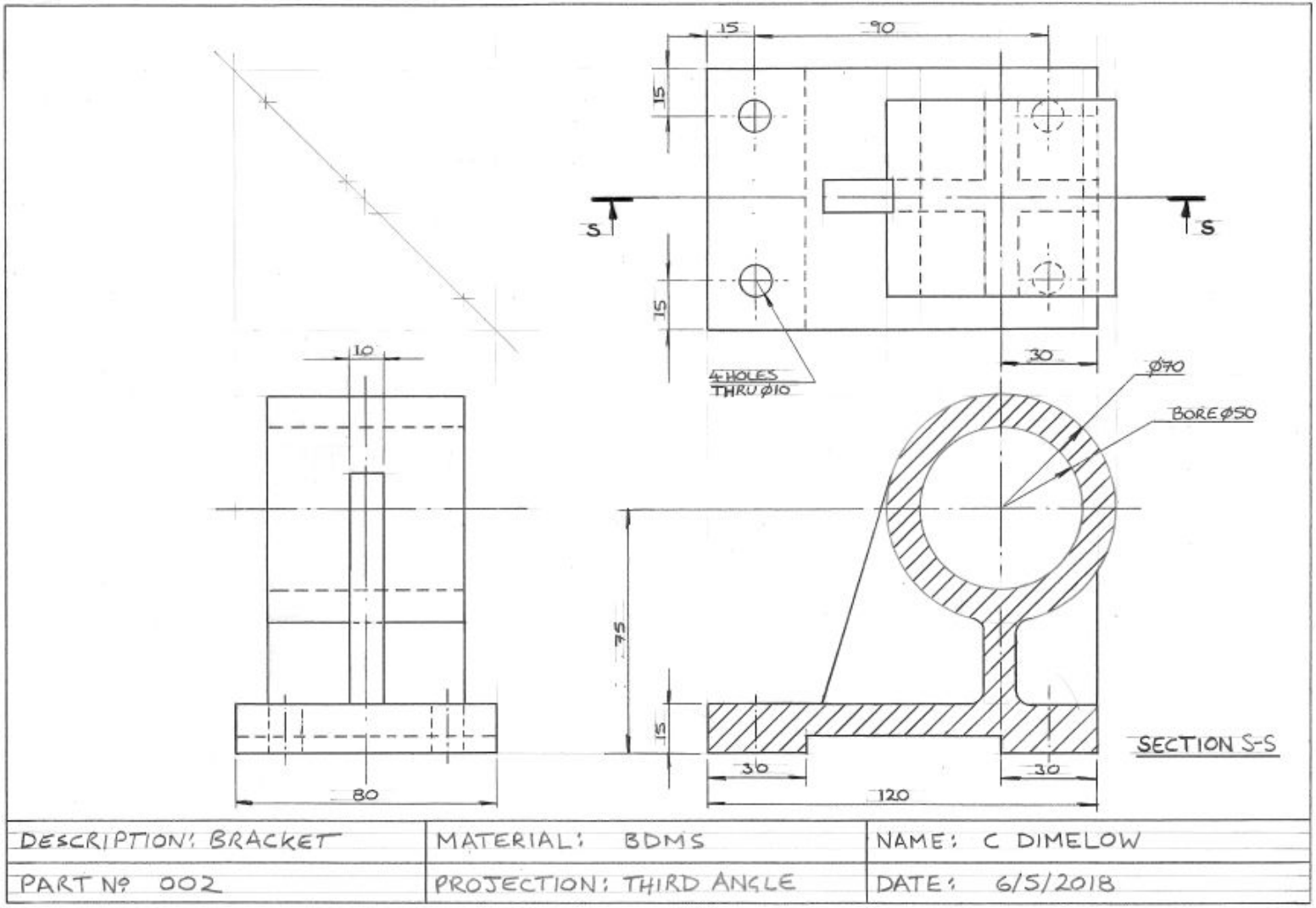
2) Project the plan view - direction 'T'

3) Project an end view - direction 'L'



Section Views

Extension 3: Solution



Orthographic Projection

Engineering Drawing

Drawing Line Types…

Line Types

There are many *line types* used in Engineering Drawing here are a few…

Visible lines

- Visible lines represent visible edges and boundaries.
- Continuous and thick (0.5 mm).

Hidden lines

- Hidden lines represent hidden edges and boundaries.
- Dashed and medium thick (0.3 mm).

Centre lines

- Centre lines Represent axes of symmetry.
- Long dash – short dash and thin (0.3 mm).

Line Types cont.

Phantom lines

- Phantom lines are used to indicate imaginary features such as alternate positions of moving parts and adjacent positions of related parts.
- The line type is long dash – short dash – short dash and the line weight is usually thin (0.3 mm).

Dimension and Extension lines

- Dimension and extension lines are used to show the size of an object. In general, a dimension line is placed between two extension lines and is terminated by arrowheads, which indicates the direction and extent of the dimension.
- The line type is continuous and the line weight is thin (0.3 mm).

Line Types cont.

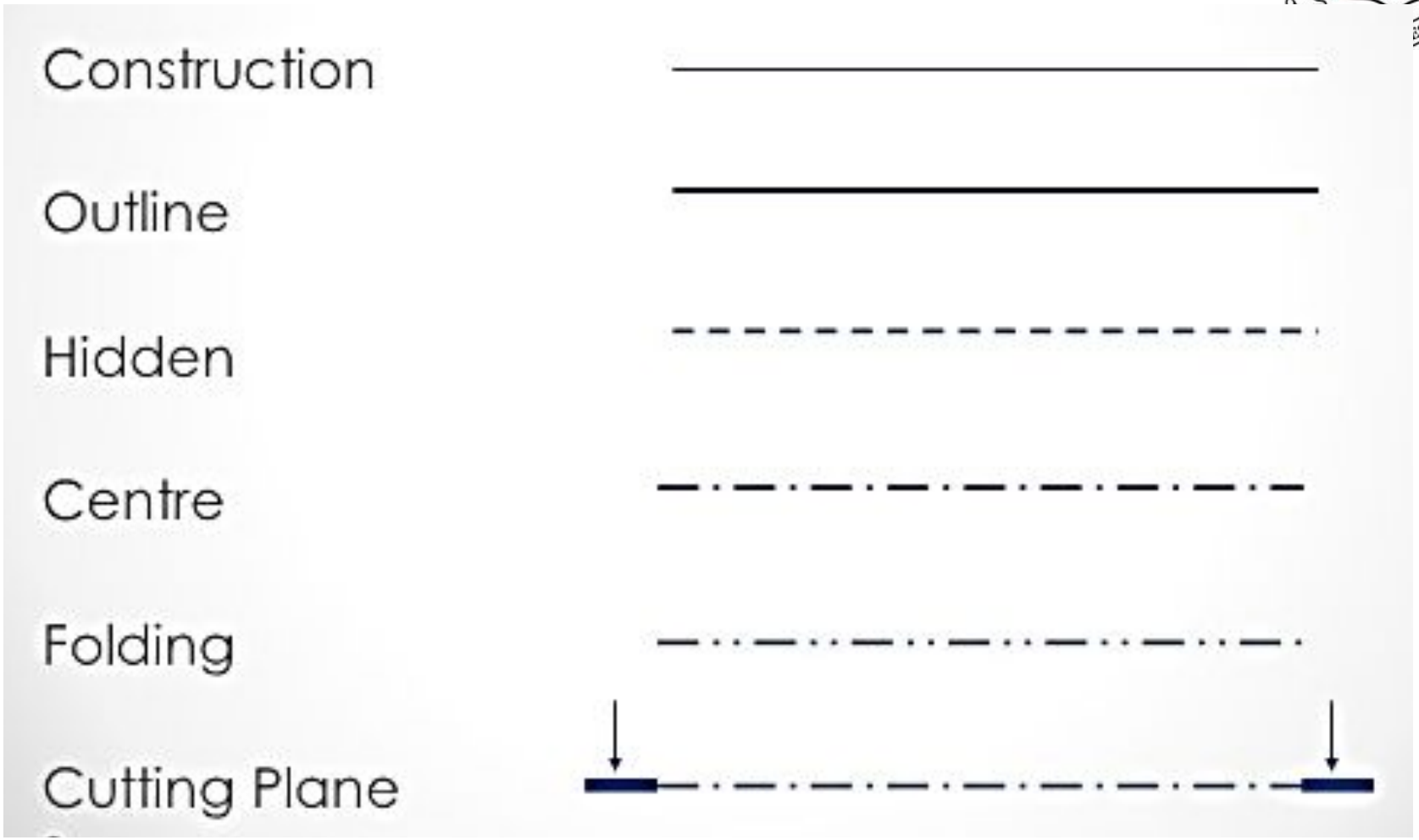
Cutting plane lines

- Cutting plane lines are used to show where an imaginary cut has been made through the object in order to view interior features.
- The line type is phantom and the line weight is very thick (0.7 mm).
- Arrows are placed at both ends of the cutting plane line to indicate the direction of sight.

Section lines

- Section lines are used to show areas that have been cut by the cutting plane.
- Section lines are grouped in parallel line patterns and usually drawn at a 45° angle.
- The line type is usually continuous and the line weight is thin (0.3 mm).

Line Types cont.



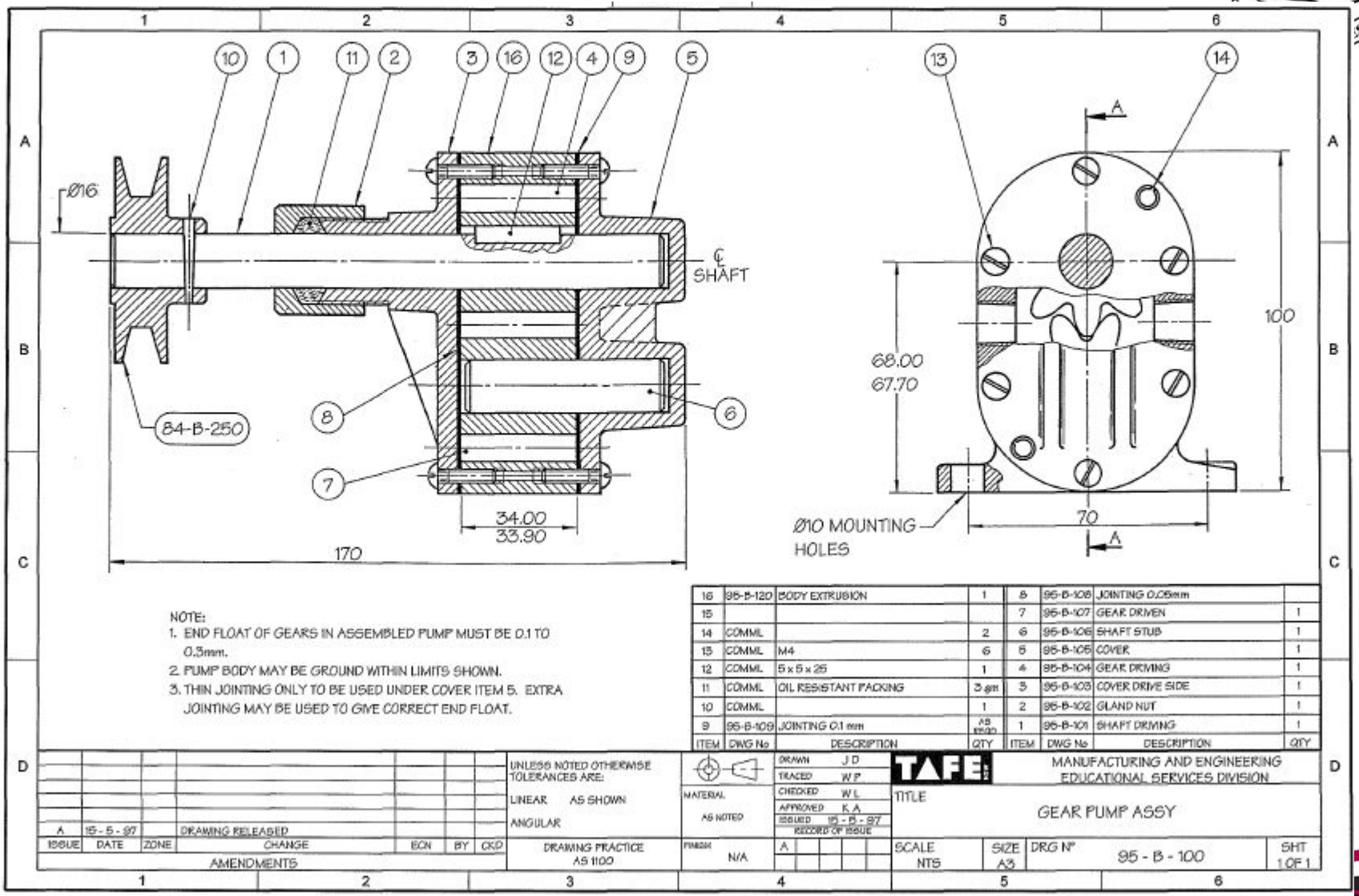
Exercise 1-3-5

Line Types - Identification

Look at the drawing ***95-B-l00*** (next slide) and answer the questions on the Line Types work sheet.

Exercise 1-3-5

Line Types - Identification



NOTE:
1. END FLOAT OF GEARS IN ASSEMBLED PUMP MUST BE 0.1 TO 0.3mm.
2. PUMP BODY MAY BE GROUND WITHIN LIMITS SHOWN.
3. THIN JOINTING ONLY TO BE USED UNDER COVER ITEM 5. EXTRA JOINTING MAY BE USED TO GIVE CORRECT END FLOAT.

ITEM	DWG No	DESCRIPTION	QTY	ITEM	DWG No	DESCRIPTION	QTY
16	95-B-120	BODY EXTRUSION	1	8	95-B-108	JOINTING 0.05mm	
15				7	95-B-107	GEAR DRIVEN	1
14	COMML		2	6	95-B-106	SHAFT STUB	1
13	COMML	M4	6	5	95-B-105	COVER	1
12	COMML	5 x 5 x 25	1	4	95-B-104	GEAR DRIVING	1
11	COMML	OIL RESISTANT PACKING	3 gm	3	95-B-103	COVER DRIVE SIDE	1
10	COMML		1	2	95-B-102	GLAND NUT	1
9	95-B-109	JOINTING 0.1 mm	AS REQD	1	95-B-101	SHAFT DRIVING	1

ISSUE	DATE	ZONE	CHANGE	ECN	BY	CKD
A	15 - 5 - 97		DRAWING RELEASED			

AMENDMENTS

UNLESS NOTED OTHERWISE TOLERANCES ARE:
LINEAR AS SHOWN
ANGULAR

MATERIAL AS NOTED

FINISH N/A

DRAWING PRACTICE AS 1100

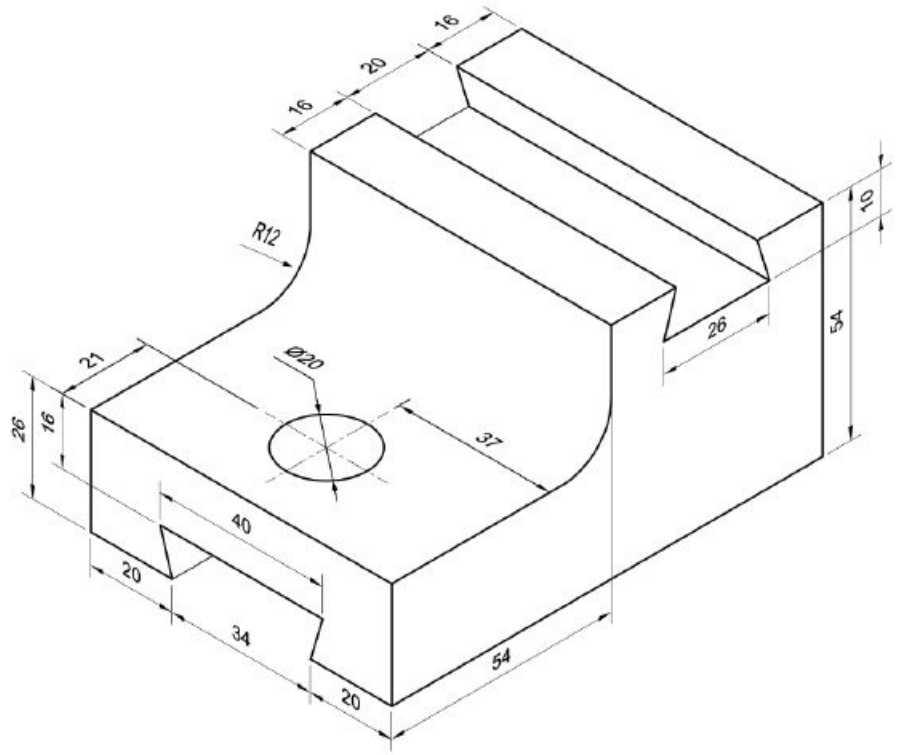
DRAWN J D
TRACED W P
CHECKED W L
APPROVED K A
ISSUED 15 - 5 - 97
RECORD OF ISSUE

TAFE
MANUFACTURING AND ENGINEERING
EDUCATIONAL SERVICES DIVISION

TITLE
GEAR PUMP ASSY

SCALE NTS | SIZE A3 | DRG N° 95 - B - 100 | SHT 1 OF 1

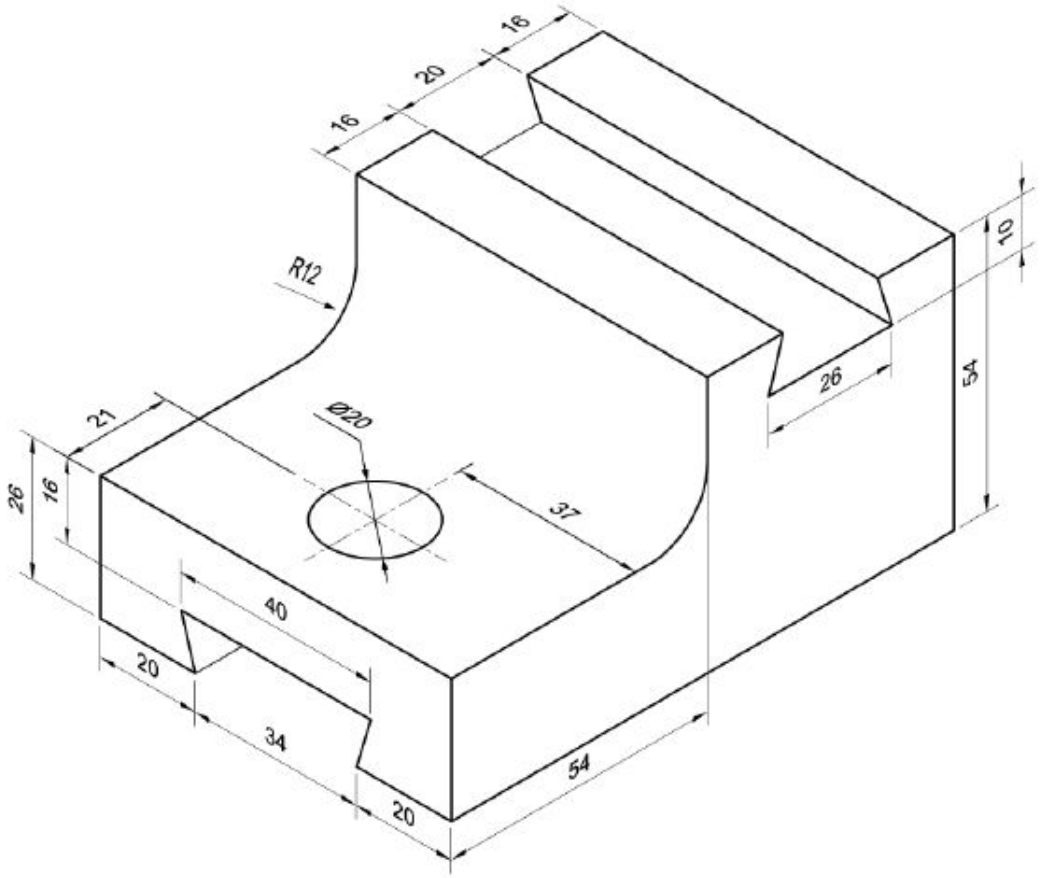
Internal Assessment 2.1 – produce a drawing



Unit 2 Assessment:

Using A3 paper and drawing equipment – add a border to the sheet and then construct the solution to the problem below.

Using the isometric drawing of the machine slide – You have ***3 hours 30 mins*** - to *construct* and *fully dimension* ***three orthographic views*** in *third angle projection*.

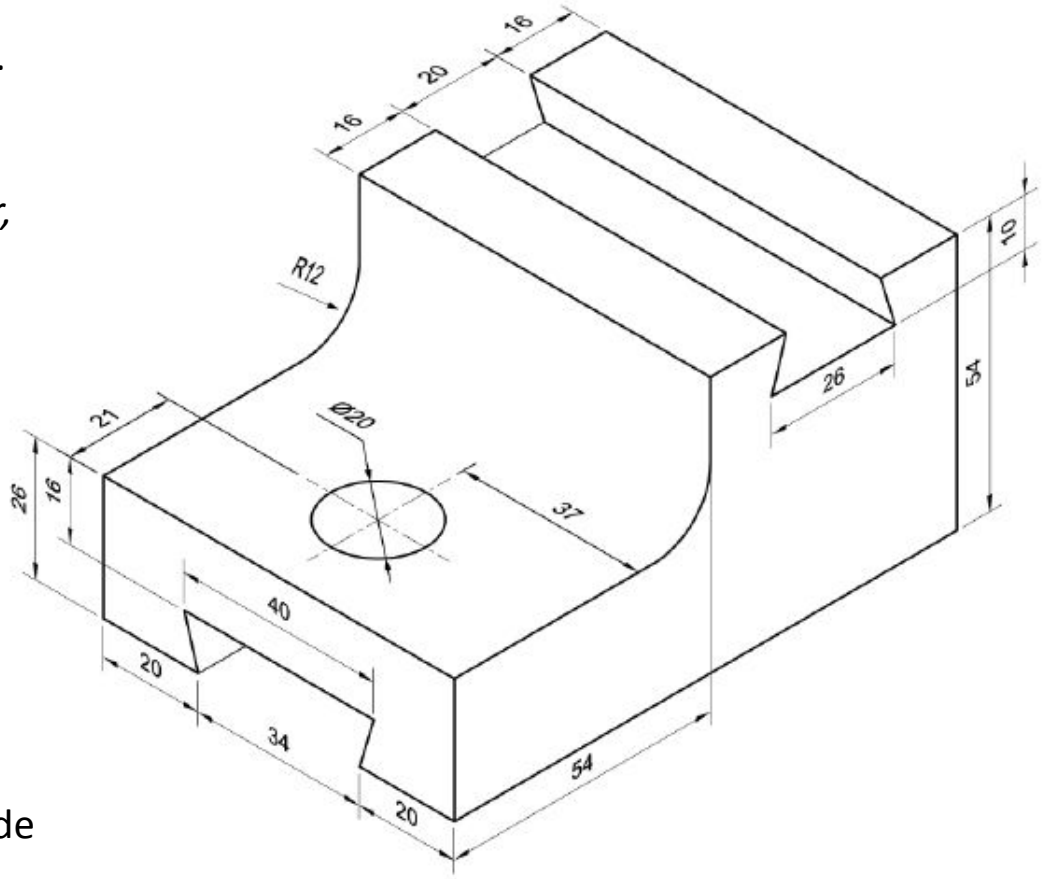


Unit 2 Assessment:

From the sketch provided produce a ***detailed*** engineering drawing in ***third angle projection***.

Create a ***border*** and ***title block*** around views, including *component description, part number, material, name of learner, date and chosen angle of projection*.

Side elevation to be a ***half section*** *lengthways* through the slide on the ***hole centre line***. ***Include*** all relevant *dimensions* and *symbols* indicating section.

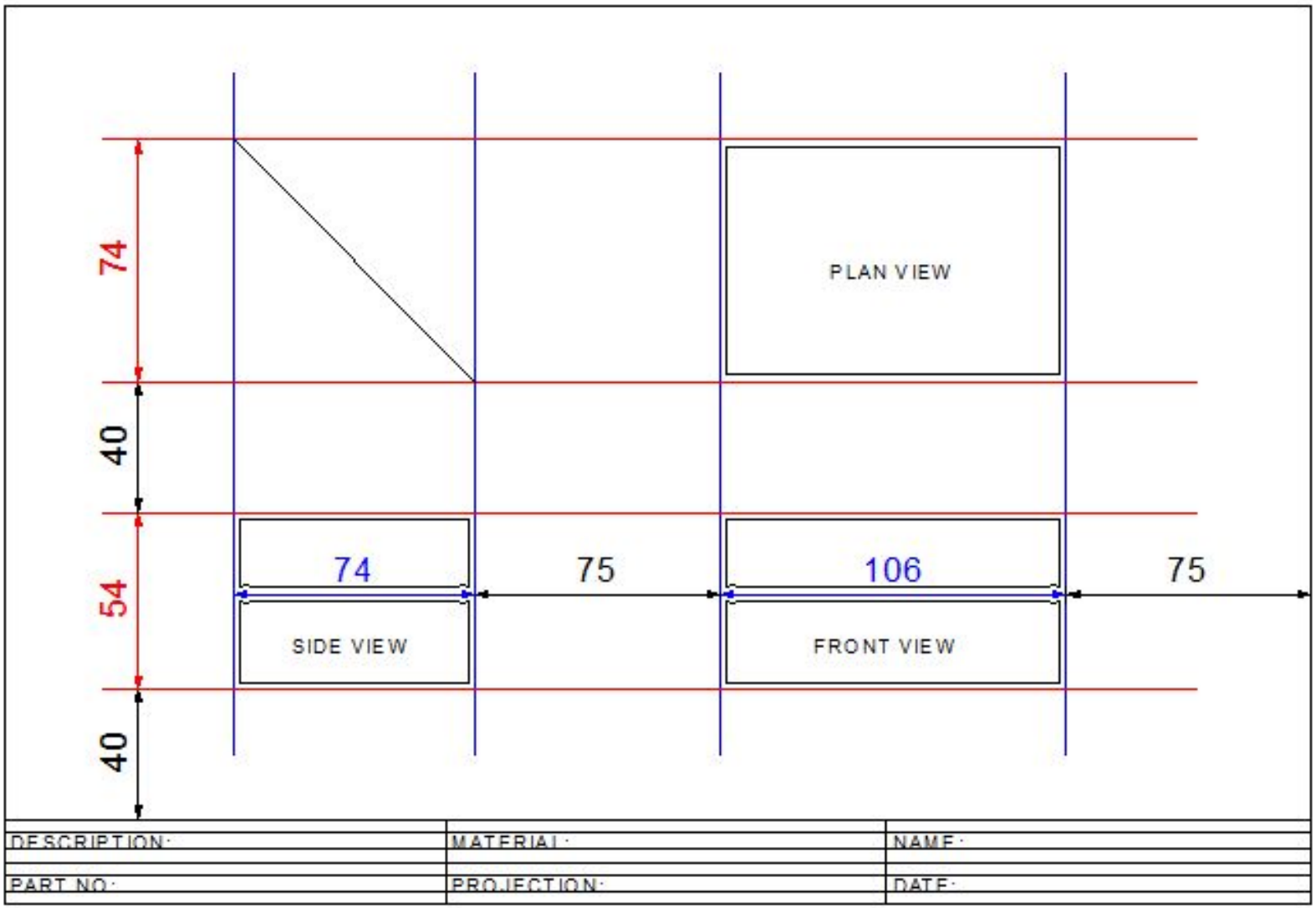


Slide
Material – Bright Mild Steel
All dimensions in mm
Note: Hole penetrates the full depth of the slide

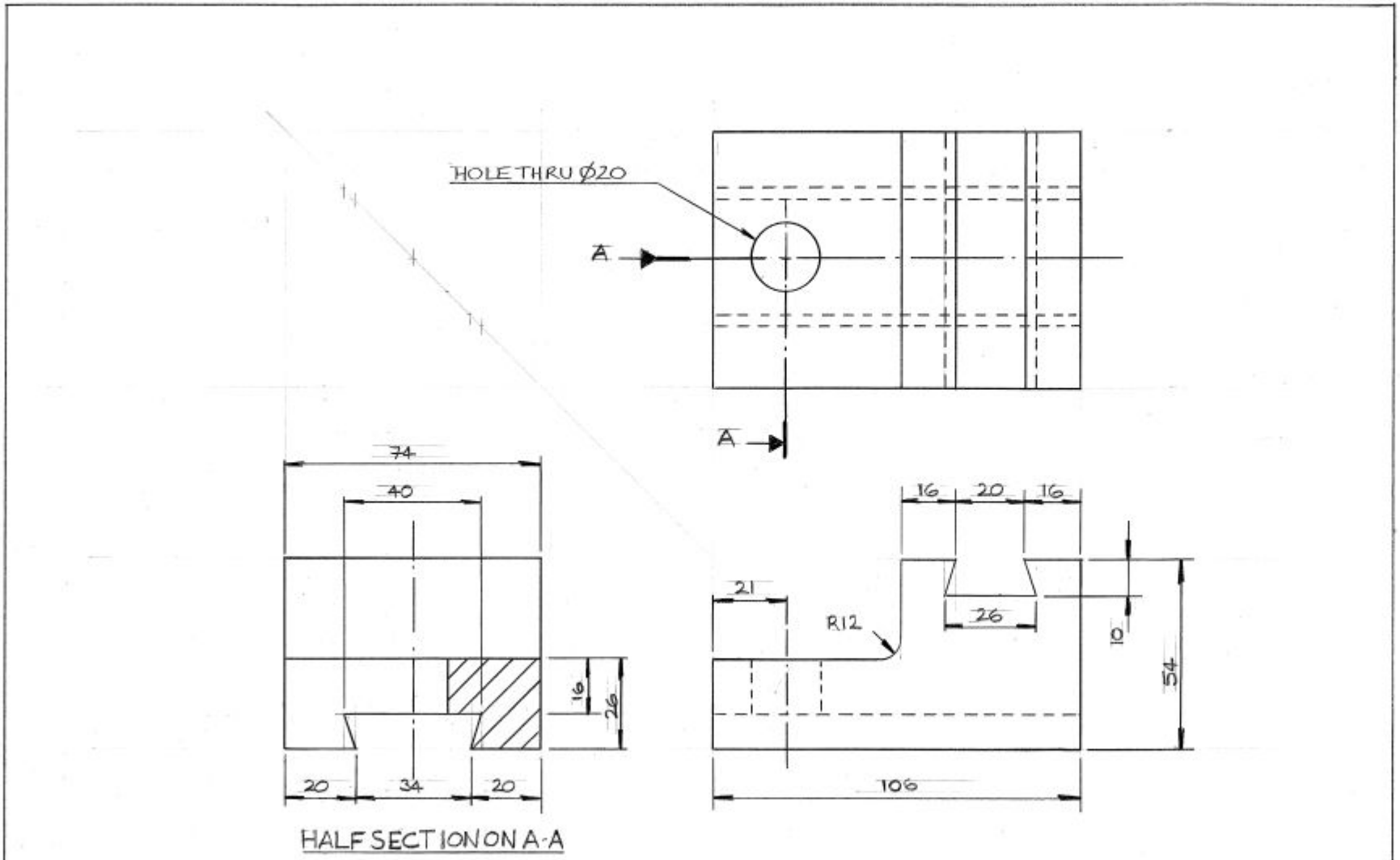
Unit 2 Assessment Mark Scheme:

Did the learner.....	Decision	
	Yes (2 marks)	No (0 marks)
Produce a drawing in three views from a given sketch		
Draw a border with a minimum of 5 title blocks labelled: Name, date, view, material and component title		
Complete the appropriate information in to each titled block		
Projection - 1st or 3rd		
End elevation view		
• Correct size		
• Lines and arrows		
Front elevation view		
• Correct size		
• Lines and arrows		
Plan elevation view		
• Correct size		
• Lines and arrows		
Section hatch		
Symbols / labels		
Sufficient detail for manufacturing purpose		

Unit 2 Assessment Set Up:



Unit 2 Assessment Solution:



DESCRIPTION: SLIDE	MATERIAL: BRIGHT MILD STEEL	NAME: C DIMELOW
PART N°: 002	PROJECTION: THIRD ANGLE	DATE: 14/07/2015